
TOMMASO MINOLA

Department of Management, Information and Production Engineering
and Center for Young and Family Enterprise
University of Bergamo

DANIELE RADICI

Innovation-LAB

INNOVAZIONE, IMPRENDITORIALITÀ E MARKETING

Course Introduction



**UNIVERSITÀ
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Annual Meeting
Davos 2025

Skills of Tomorrow

10 00:03 / 00:52

CC

<https://www.weforum.org/videos/what-are-the-most-essential-skills-in-the-workplace-of-tomorrow/>

What is entrepreneurship for you?



What is entrepreneurship for you? (Babson College)

"Is **skydiving while constructing** your parachute."

Daniel (Friend of Babson, US)

"Is having the **creativity**, knowledge and skills to make an idea turn into reality."

Shoney
(Undergrad, Uzbekistan)

"Is being able to **see** what others can only imagine."

Andrea
(Undergraduate student, US)

"Is **making impossible POSSIBLE.**"

Rajesh (Babson Alumni, India)

"Means **not being afraid** to fail ... and not giving up after you do."

Meredith (Babson Alumni, MA)



“One of the only ways **to get out of a tight box** is to invent your way out.”

Jeff Bezos

“If we tried to think of a good idea, we wouldn't have been able to think of a good idea at all. **You just have to have a solution for a problem in your own life.**”

Brian Chesky



“AI won't take your job, but someone else **who uses AI might**”

Jensen Huang

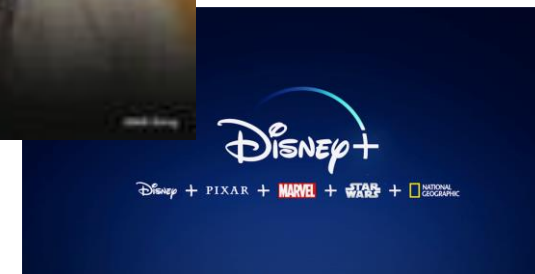
“If you've got an idea, **start today.**
There's **no better time than now** to get going”

Kevin Systrom





https://youtu.be/Y4ZvI2u_57E?si=U1QW-0NZyjYZbVRF





<https://www.youtube.com/watch?v=ZUG9qYTJMsI>



DOLLAR SHAVE CLUB



<https://youtu.be/d7eRafWqqRg>

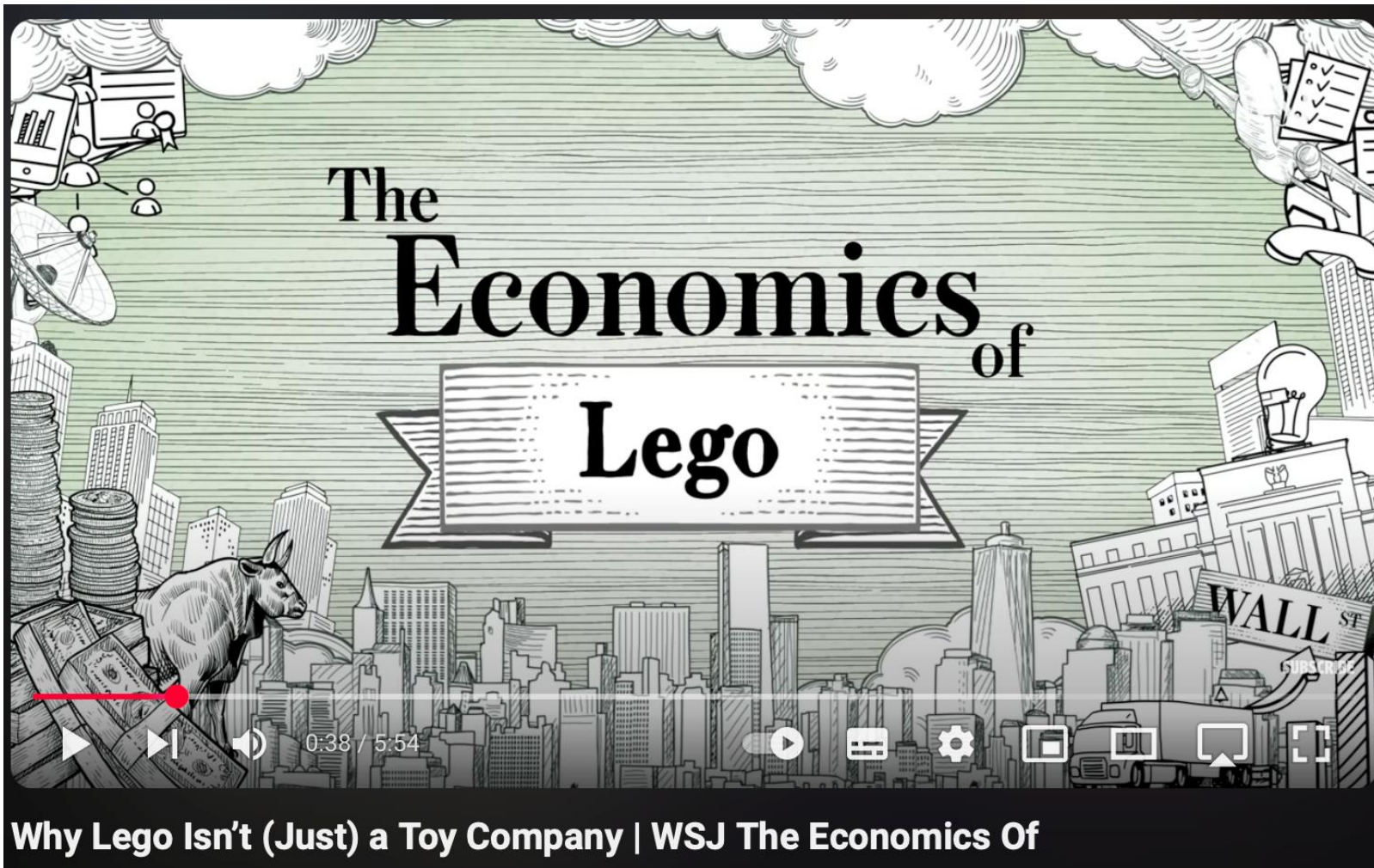


Postmates

Fleet Management From Hilti - How Does It Work?



<https://youtu.be/CNKccALnZWY?feature=shared>



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<https://www.youtube.com/watch?v=mooVZ8Uh4bo>

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Mutual

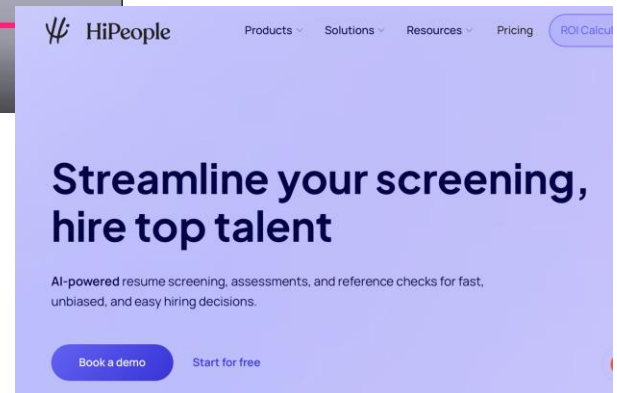
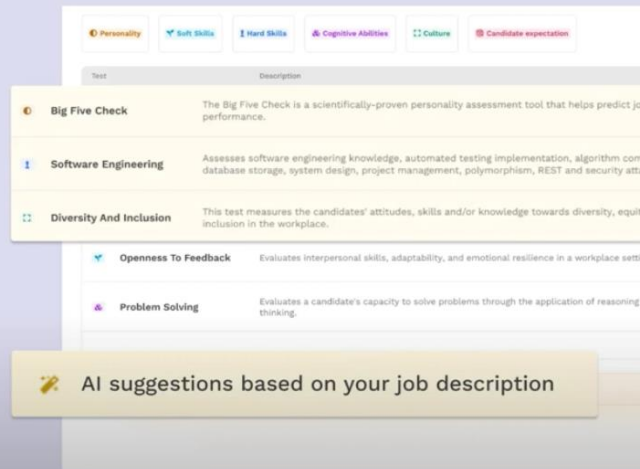
Pearson



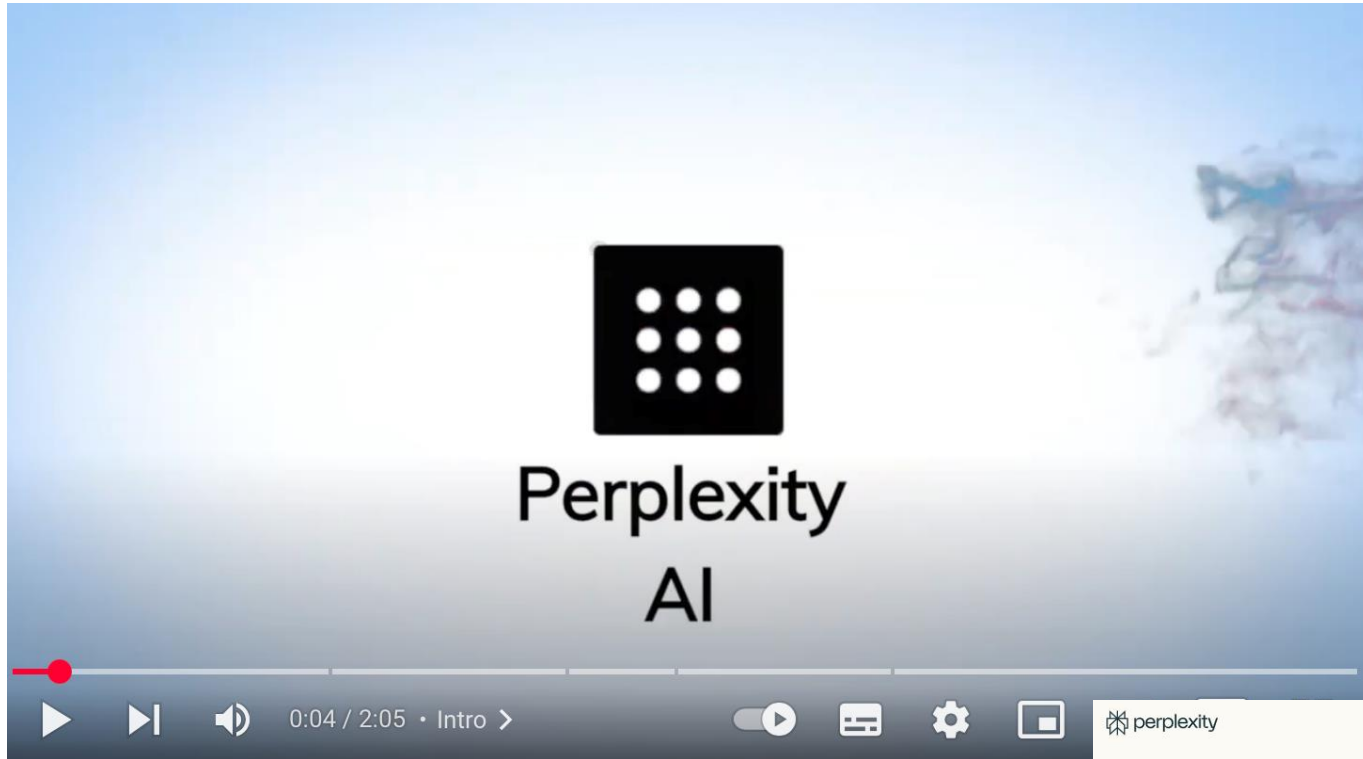
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HiPeople AI

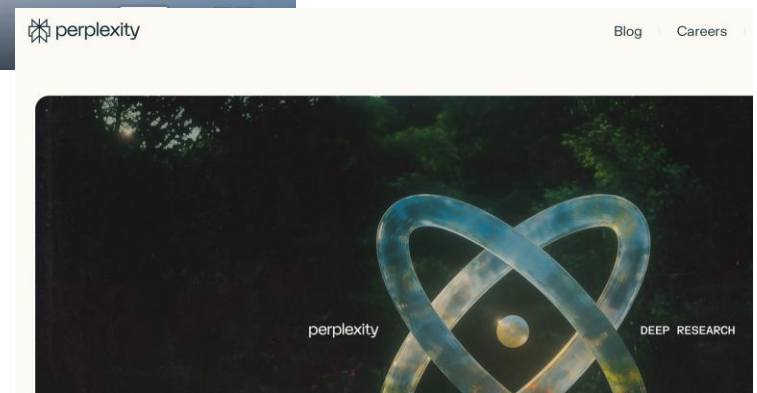
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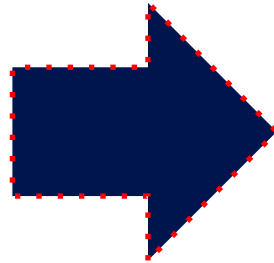
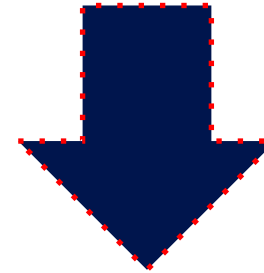


<https://youtu.be/4sMKmrjxHdA?feature=shared>



Exercise 1: What are your thoughts?

Please discuss in small groups the **common characteristics** of these start-ups and share your answers:



- **A world of Unicorns?**

Unicorns, Zabras, Camels, Gazelles, Black swans etc.

Entrepreneurial species*	Suggested definition	Examples
 Unicorn	Unicorns are ventures valued at more than one billion dollars in the first years of life (Lee, 2013)	Uber, SpaceX
 Gazelle	Gazelles are ventures under five years old with an average employment growth rate of more than 20% per year over three years, and with ten or more employees at the beginning of each reporting period (Organization for Economic Cooperation and Development; Ahmad, 2006)	Cisco, Yahoo
 Zebra	Zebras are ventures that prioritize both profitability and societal impact. They emphasize mutualistic relationships, are resilient, and use alternative fundraising methods. They champion steady growth, worker welfare, and sustainability (Brandel <i>et al.</i> , 2017; Zepeda, 2017; de Gennaro <i>et al.</i> , 2023)	Switchboard, Hearken
 Camel*	Camels are resilient ventures that survive in tough conditions. These ventures are prepared for the long haul, often patiently building their business model and awaiting the right moment for major growth. Prioritizing balanced growth and long-term sustainability over rapid expansion, they emphasize proper pricing, prudent cost management, and flexibility in their growth plans. Camels often diversify by targeting multiple markets and constructing a comprehensive range of products and services to bolster resilience (Crunchbase, 2020; Lazarov, 2021; Forbes, 2023)	Grubhub, Zooma
 Donkey*	Donkeys are ventures that began as high-potential unicorns but failed to justify their lofty valuations. Often, they prioritize aggressive marketing over critical business facets and may neglect basic business economics (e.g., product margins). In some cases, they face challenges during their transition from private to public entities, leading to significant financial or reputational losses (German entrepreneurship, 2021; Gov.ph, 2022)	WeWork
 Mouse*	Mouse ventures are small businesses with up to 20 employees, often specializing in niche markets or social missions. While they can quickly adapt and innovate, their size makes them vulnerable to larger competitors and market changes. Although their macroeconomic impact may be limited, they play a significant role in market dynamics and social innovation (Birch, 1987; Acs and Muller, 2008; Austin and Leonard, 2008; Waddock, 2008)	Local artisanal shops, freelance consultants, etc.

Unicorns, Zabras, Camels, Gazelles, Black swans etc.

 Elephant	Elephant ventures are large, established corporations that, due to their size and structural complexity, often move and adapt slowly. Their large-scale grants them significant economic and organizational resources. However, this same scale can make them bureaucratic and risk-averse (Birch, 1987; Acs and Muller, 2008; Austin and Leonard, 2008; Waddock, 2008)	Colgate, Coca-Cola, Unilever
 Gorilla	Gorilla ventures are dominant in their industry, boasting substantial market shares without reaching monopoly status. Their significant presence grants them the luxury of taking risks without jeopardizing their customer base, as their size offers a cushion against potential setbacks. These ventures serve as influential market leaders, setting trends and standards. Their magnitude often enables collaborations with smaller ventures, seeking innovation and maintaining a positive corporate image (Prashantham and Birkinshaw, 2008; Prashantham, 2021)	McDonalds, Microsoft
 Dragon	Dragon ventures refer to those valued at \$12 billion or more, net of venture funding. These ventures not only possess substantial financial worth but also provide returns that cover an entire venture fund, making them highly valuable for investors. Dragons emphasize genuine long-term value and resilience, prioritizing sustainable business models over mere valuation figures (Axios, 2021; Fast Company, 2023)	Stripe, ByteDance
 Lame Duck	Lame duck are ventures that have seen a significant decline in their influence, market presence, or profitability, or ones which have defaulted on their obligations or gone bankrupt due to trading losses, rendering them ineffective in the market (Goldsmith, 2009; Investopedia, 2022)	Blockbuster, Blackberry
 Phoenix	Phoenix ventures arise when the business of an insolvent venture is transferred to a new entity, effectively allowing it to continue operations, while leaving behind its debts. This practice, while legal under specific conditions, is controversial due to potential misuse and fraud (Experian, 2022; Gov.uk 2023)	Alitalia-ITA Airways; Morris Garage

Note(s): *entrepreneurial evolutions not investigated within Special Issue articles

Source(s): Own elaboration

Black swans are rare events, especially those that are random and unexpected.

Aldrich, H. E., & Ruef, M. (2018). Unicorns, gazelles, and other distractions on the way to understanding real entrepreneurship in the United States. *Academy of Management Perspectives*, 32(4), 458-472.

Cristofaro, M., Kask, J., & Muldoon, J. (2023). Guest editorial: Exploring the entrepreneurial jungle: unicorns, gazelles, zebras and other venture species. *Journal of Small Business and Enterprise Development*, 30(6), 1065–1087. <https://doi.org/10.1108/JSBED-10-2023-509>

Unicorns represent a growth paradigm in contemporary technological entrepreneurship that is centered on rapid expansion and capital-intensive scaling.

Main characteristics

- Pursuit of **hyper-fast, exponential growth**
Strong reliance on **venture capital and external funding**
- Focus on **global scalability and billion-dollar valuations**
- Adoption of a “**growth before profitability**” strategy
- Emphasis on **rapid market penetration and network effects**

Zabras are interpreted as an **ethical and sustainable response to the unicorn growth model**.

Main characteristics

- Emphasis on **sustainable growth** rather than extreme expansion
- Simultaneous pursuit of **economic returns and social impact**
- Strong focus on **stakeholders**, rather than exclusively on shareholder value
- **Long-term orientation and organizational stability**
- Growth strategies based on **balance, responsibility, and moderation**

Camels represent a growth logic **centered on resilience and long-term survival**.

Main characteristics

- Strong focus on **cash-flow management and self-financing**
- Limited dependence on external capital
- Emphasis on **risk control and financial prudence**
- Ability to survive and operate effectively in unstable or crisis-prone environments
- **Slower growth pace, but greater structural solidity**

Gazelles are defined as **small and medium-sized enterprises that sustain relatively high growth rates over time**, playing a significant role within the economic system.

Main characteristics

- Annual growth rates typically around **20–25%**
- Growth achieved without excessive capital intensity
- Maintenance of organizational and financial discipline during expansion
- Significant contribution to **job creation and local economic development**
- Two growth rates: unique venture-specific characteristics (single innovative products) and external factors (usually connected with the environment in which they are embedded)



STARTUPS



More than 100 new tech unicorns were minted in 2025 — here they are

<https://techcrunch.com/2026/01/12/at-least-36-new-tech-unicorns-were-minted-in-2025-so-far/>

Dominic-Madori Davis — 5:00 PM PST • January 12, 2026

luma*

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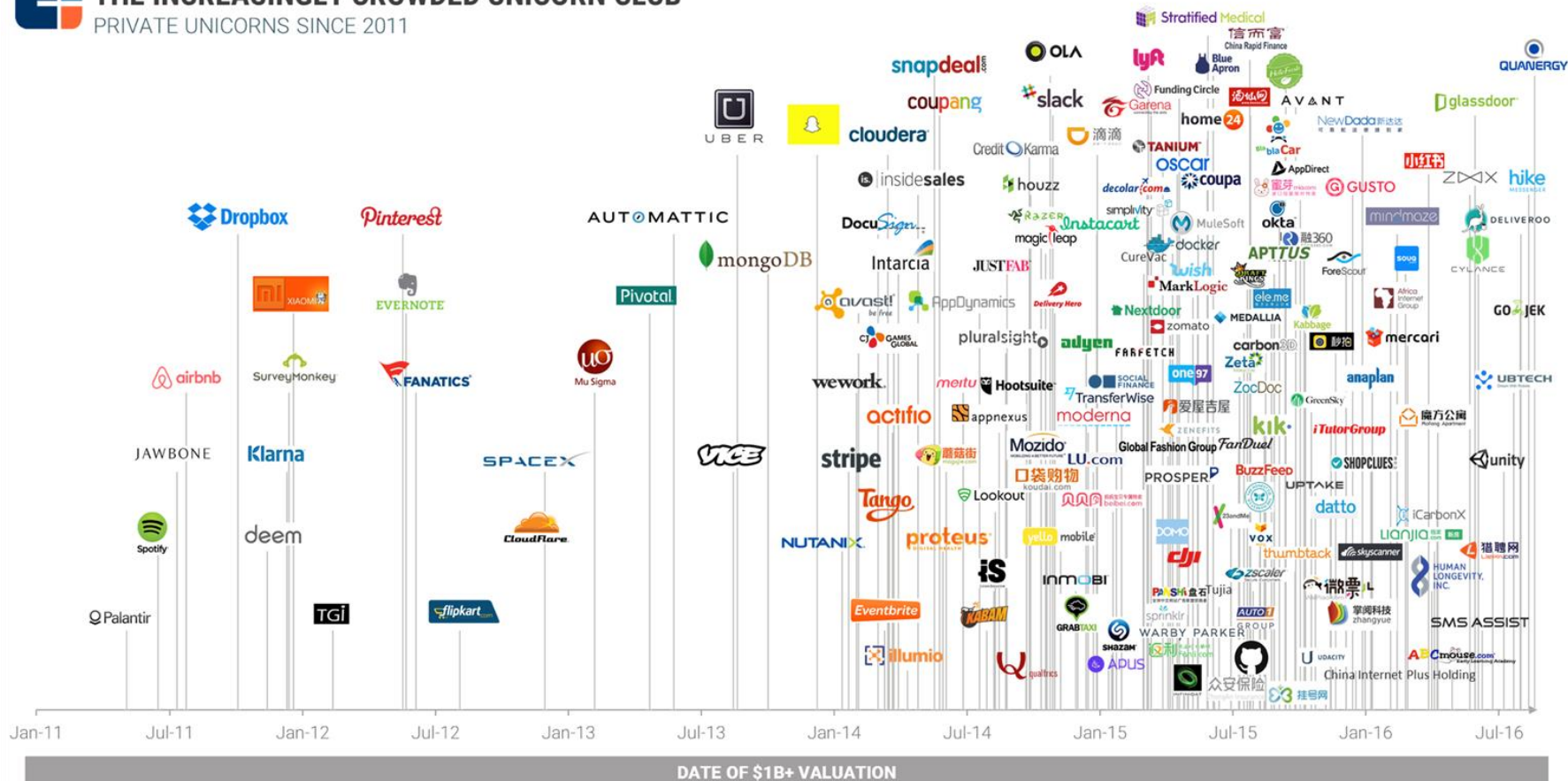
GET STARTED → TALK TO OUR TEAM

Category	Definition	Number of Companies (Approx.)	Typical Company Names
AI-related unicorns	AI is the core product or business model (foundation models, generative AI, AI agents, AI infrastructure)	62 (53%)	Thinking Machines, Reflection, Luma, Suno, Fireworks AI, LangChain, Baseten, Decagon
Non-AI unicorns	Core business does not rely on AI (energy, aerospace, blockchain, consumer goods, hardware, traditional SaaS)	46 (39%)	Kalshi, Loft Orbital, Radiant, Oishii, Stoke, Polymarket, Olipop, Substack
Hybrid / AI-enabled unicorns	AI is used as an enabling feature, but not the primary value proposition	10(8%)	You.com, Linear, Hightouch, ThreatLocker, Teamworks
Total	118		

<https://techcrunch.com/2026/01/12/at-least-36-new-tech-unicorns-were-minted-in-2025-so-far/>

THE INCREASINGLY CROWDED UNICORN CLUB

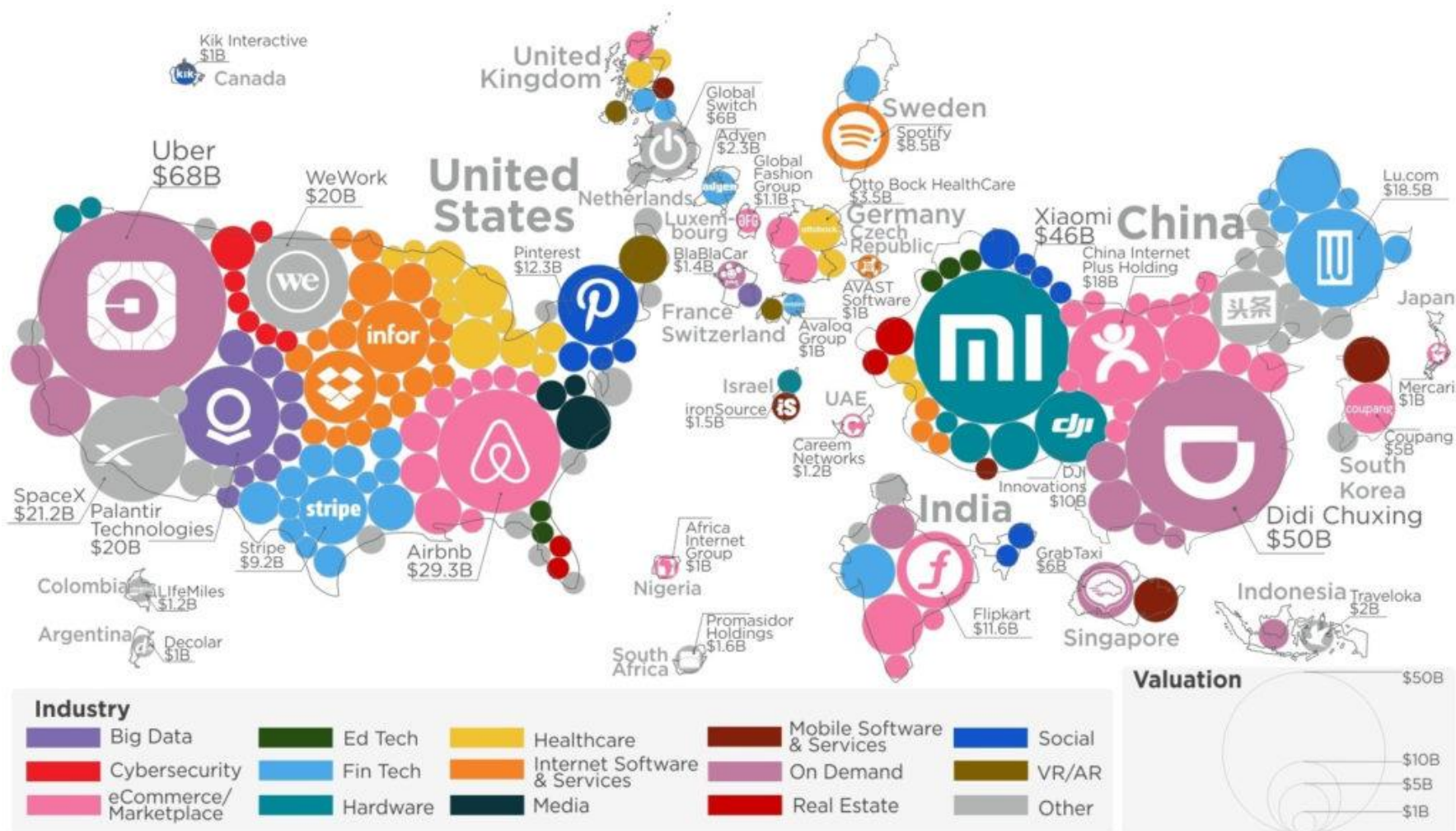
PRIVATE UNICORNS SINCE 2011



www.cbinsights.com

*excludes companies that exited

CBINSIGHTS



howmuch.net

List of unicorn companies in November, 2025

Company	Valuation (\$B)	Date Joined	Country	City	Industry	Select Investors	Company	Valuation (\$B)	Date Joined	Country	City	Industry	Select Investors
SpaceX	\$350	12/1/2012	United States	Hawthorne	Industrials	Founders Fund, Draper Fisher Jurvetson, Rothenberg Ventures	OpenAI	\$500	7/22/2019	United States	San Francisco	Enterprise Tech	Khosla Ventures, Thrive Capital, Sequoia Capital
ByteDance	\$300	4/7/2017	China	Beijing	Media & Entertainment	Sequoia Capital China, SIG Asia Investments, Sina Weibo, SoftBank Group	ByteDance	\$480	4/7/2017	China	Beijing	Media & Entertainment	Sequoia Capital China, SIG Asia Investments, Sina Weibo, SoftBank Group
OpenAI	\$157	7/22/2019	United States	San Francisco	Enterprise Tech	Khosla Ventures, Thrive Capital, Sequoia Capital	SpaceX	\$400	12/1/2012	United States	Hawthorne	Industrials	Founders Fund, Draper Fisher Jurvetson, Rothenberg Ventures
Stripe	\$70	1/23/2014	United States	San Francisco	Financial Services	Khosla Ventures, LowercaseCapital, capitalG	Anthropic	\$350	2/3/2023	United States	San Francisco	Enterprise Tech	Google
SHEIN	\$66	7/3/2018	Singapore		Consumer & Retail	Tiger Global Management, Sequoia Capital China, Shunwei Capital Partners	Databricks	\$100	2/5/2019	United States	San Francisco	Enterprise Tech	Andreessen Horowitz, New Enterprise Associates, Battery Ventures
Databricks	\$62	2/5/2019	United States	San Francisco	Enterprise Tech	Andreessen Horowitz, New Enterprise Associates, Battery Ventures	Revolut	\$75	4/26/2018	United Kingdom	London	Financial Services	Index Ventures, DST Global, Ribbit Capital
Anthropic	\$60	2/3/2023	United States	San Francisco	Enterprise Tech	Google	Stripe	\$70	1/23/2014	United States	San Francisco	Financial Services	Khosla Ventures, LowercaseCapital, capitalG
xAI	\$50	5/26/2024	United States	Burlingame	Enterprise Tech	Sequoia Capital, Andreessen Horowitz, VY Capital	SHEIN	\$66	7/3/2018	Singapore		Consumer & Retail	Tiger Global Management, Sequoia Capital China, Shunwei Capital Partners
Revolut	\$45	4/26/2018	United Kingdom	London	Financial Services	Index Ventures, DST Global, Ribbit Capital	xAI	\$50	5/26/2024	United States	Burlingame	Enterprise Tech	Sequoia Capital, Andreessen Horowitz, VY Capital
Canva	\$32	1/8/2018	Australia	Surry Hills	Enterprise Tech	Sequoia Capital China, Blackbird Ventures, Matrix Partners	Ripple	\$40	12/20/2019	United States	San Francisco	Financial Services	IDG Capital, Venture51, Lightspeed Venture Partners
							Figure	\$39	2/23/2024	United States	Sunnyvale	Industrials	Intel Capital, Parkway VC, Amazon Industrial Innovation Fund
							Ramp	\$32	3/29/2021	United States	New York	Enterprise Tech	D1 Capital Partners, Stripe, Coatue Management
							Canva	\$32	1/8/2018	Australia	Surry Hills	Enterprise Tech	Sequoia Capital China, Blackbird Ventures, Matrix Partners
							Fanatics	\$31	6/6/2012	United States	Jacksonville	Consumer & Retail	SoftBank Group, Andreessen Horowitz, Temasek Holdings

- Total number of unicorn companies worldwide: 1334
- Total cumulative valuation: ~\$5,443 billion



What is it?

- Currently based in Milan.
- Operates in the FinTech market and provides new ways to shop.
- For businesses: Allow your customers to pay in instalments for their order through ScalaPay.
- For customers: Able to pay on instalments for thousands of products in hundreds of retailers.

<https://www.scalapay.com/it>

Story:

- 2019 - Launch **in Italy**.
- 2021 - Launch in **the rest of Europe**; \$40m seed round investment; Launch of **Pay by Link**.
- 2022 - **Unicorn status \$1B valuation**; Partnership with **Twig** and **Web3** green payment to empower the circular economy; to provide Gen-Z and Millennials with a sustainable shopping experience.
- 2023 - Series B round that netted ScalaPay \$497 Million to **further its aggressive expansion** and internal growth.

- A world of Unicorns?

Unicorns are emerging at an accelerating pace, especially in the U.S. and Asia. Most are rooted in technology and digital innovation, reshaping industries. However, their true economic impact remains a subject of debate—are they genuine drivers of growth or just an overhyped myth?

- **Changing the world... for good!**

Functional role of technology in health (pre-AI focus)

- Technology is used to:
 - manage health-related information
 - support prevention and well-being
 - enable continuous interaction between users and services
- Health innovation focuses on:
 - prevention rather than acute treatment
 - long-term well-being

Storage & transmission
Information support
Descriptive



AI as a technological driver in healthcare

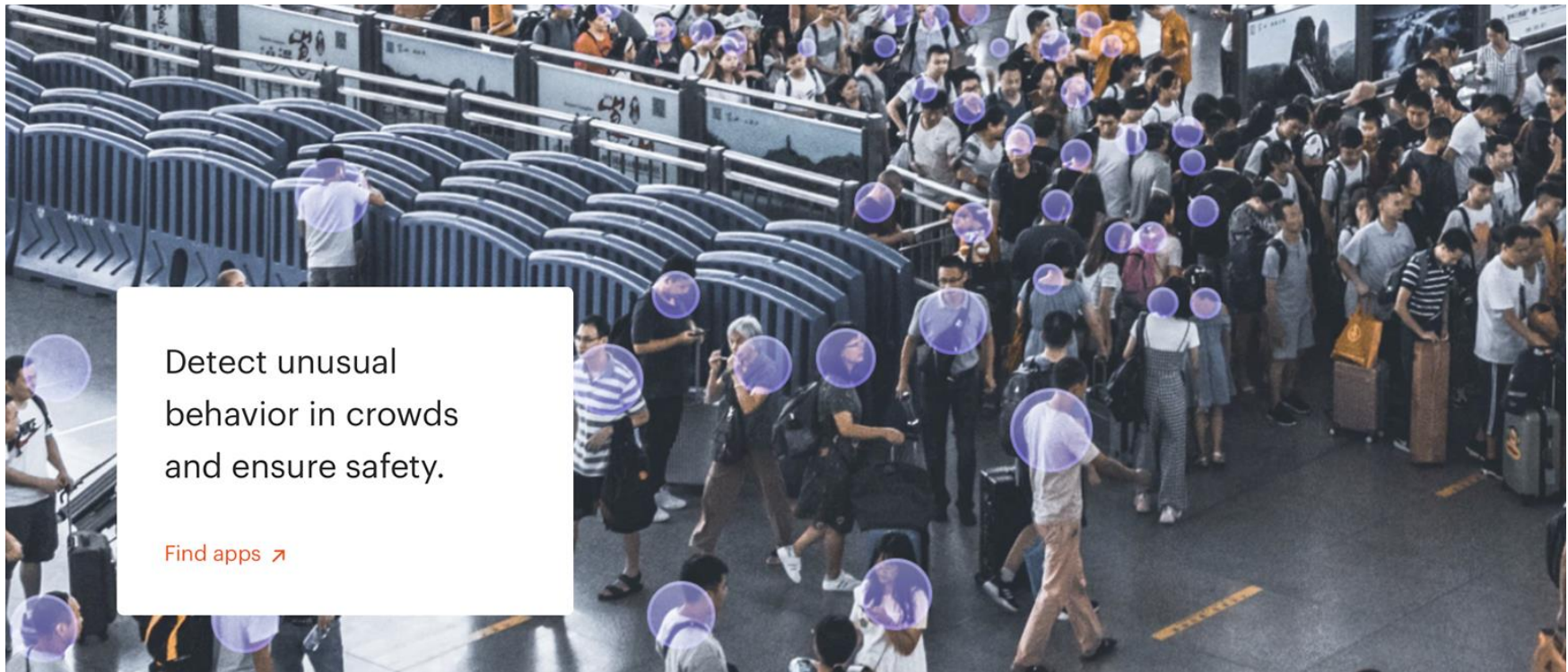
- Artificial intelligence is already technically capable of:
 - supporting diagnosis
 - risk prediction
 - decision support
- AI healthcare solutions are:
 - data-driven
 - algorithm-based
 - increasingly specialized

Analysis & prediction
Decision support
Predictive

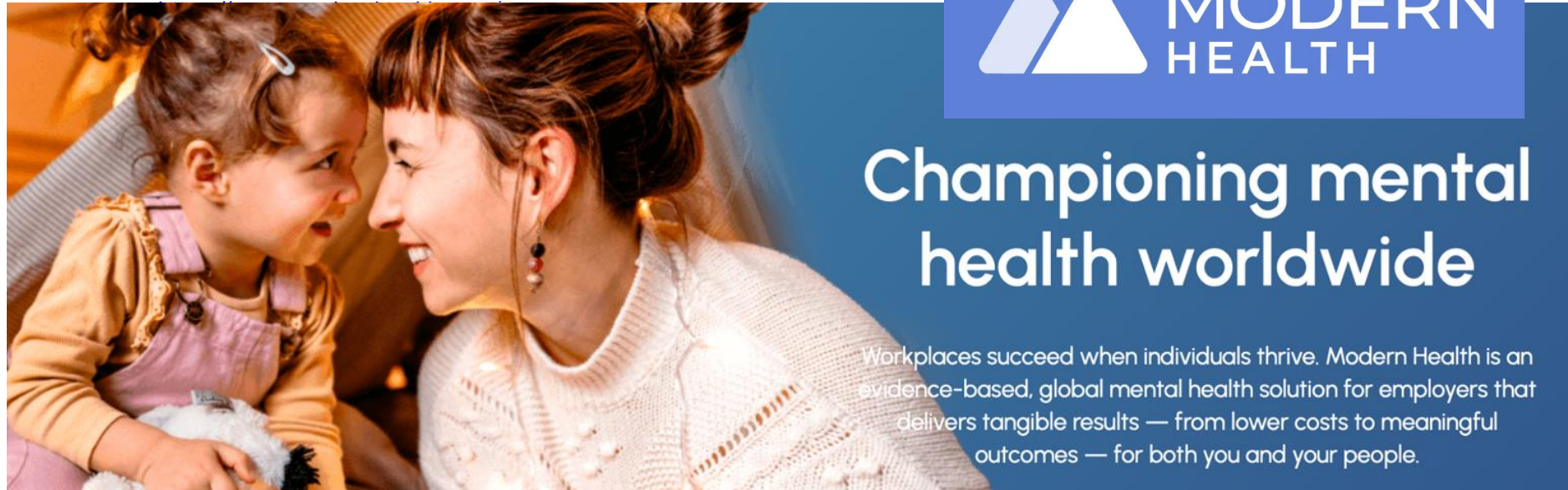




S&ST is in the security-camera business, a vertically integrated industry with a history of slow software innovation. S&ST disrupted the industry by building an eco-system to provide an open platform where software developers can build apps to run on many different types of security cameras. **The platform went live in the middle of the pandemic.** S&ST is a Robert Bosch corporate start-up.



Source: "Medtech innovation in the time of COVID-19: Resilience and growth during a crisis", McKinsey Digital, Oct 2020



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- Founded in 2017 by Alyson Friedensohn Watson and Erica Johnson in San Francisco, California.
- Operates a B2B2C model, offering solutions for organizations including employers, providers, small & medium businesses, and large enterprises.
- Its apps packages several forms of care for:
 - Mental disorder to fight serious illnesses,
 - Mental self-improvement to resolve stress or anxiety.

<https://research.contrary.com/reports/modern-health>



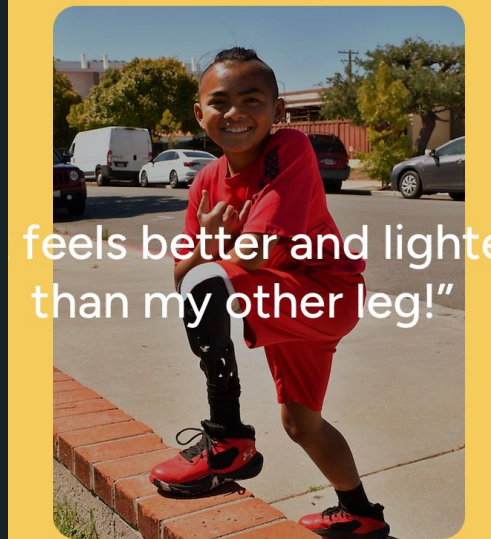
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<https://www.limberprosthetics.com/>



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<https://www.polihub.it/news/bi-rex-vince-ledizione-2020-di-startup4good/>



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- **Access to justice using natural language models and machine learning**
(Gerald Abila, Founder, Barefoot Law, Uganda)
- In a nation with one solicitor to twenty-thousand individuals, obtaining answers to fundamental and legal queries can imply survival or demise, and authority or disempowerment.
- The company deploys an AI solution named Winnie, an AI-lawyer that provides pro-bono legal services to Ugandans, especially citizens of rural communities who would not otherwise access legal redress.
- To date, Barefoot Law's solution has facilitated over 800,000 users to access their services and resolved over 20,000 cases.

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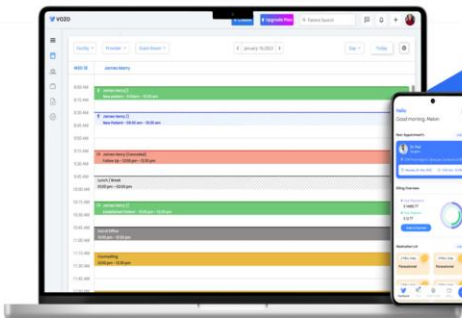
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Refugees



Climate
advocates



U.S.
entrepreneurs

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<https://www.kiva.org/>

-Kiva ([kiva.org](https://www.kiva.org)) is an international nonprofit with a mission to expand financial access to help underserved communities thrive.

-They run a global marketplace platform to crowdfund microloans for financially excluded entrepreneurs, farmers and students around the world.

-Their organization combines the culture and technological passion of an internet start-up with the compassion and empathy of a non-profit to create impact and opportunity at global scale.

-In just 13 years, they have raised more than \$1.2 billion in loan capital for 2.7 million borrowers in 83 countries



SUSTAINABLE DEVELOPMENT GOALS

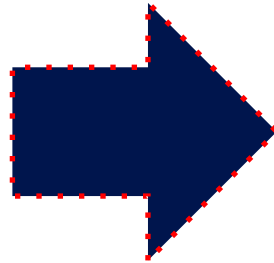
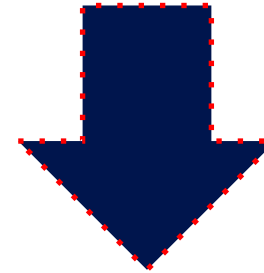


An example from my own research:

https://drive.google.com/file/d/1RPpolhbvtIA2QRXJnnJ1m9b9CvI_7LAj/view?usp=sharing

Exercise 2: What are your thoughts?

Please list in small groups **(digital) startups that you now, addressing SDGs** and share your answers:



- A world of Unicorns?
- Changing the world... for good!

Entrepreneurs are reshaping the world, using technology to uplift vulnerable communities, protect the environment, and revolutionize health, education, finance, and many other industries. The SDG framework highlights where innovation is driving change—but the future is still being written. Where will you make your impact?

- **No *playground* for old men?**







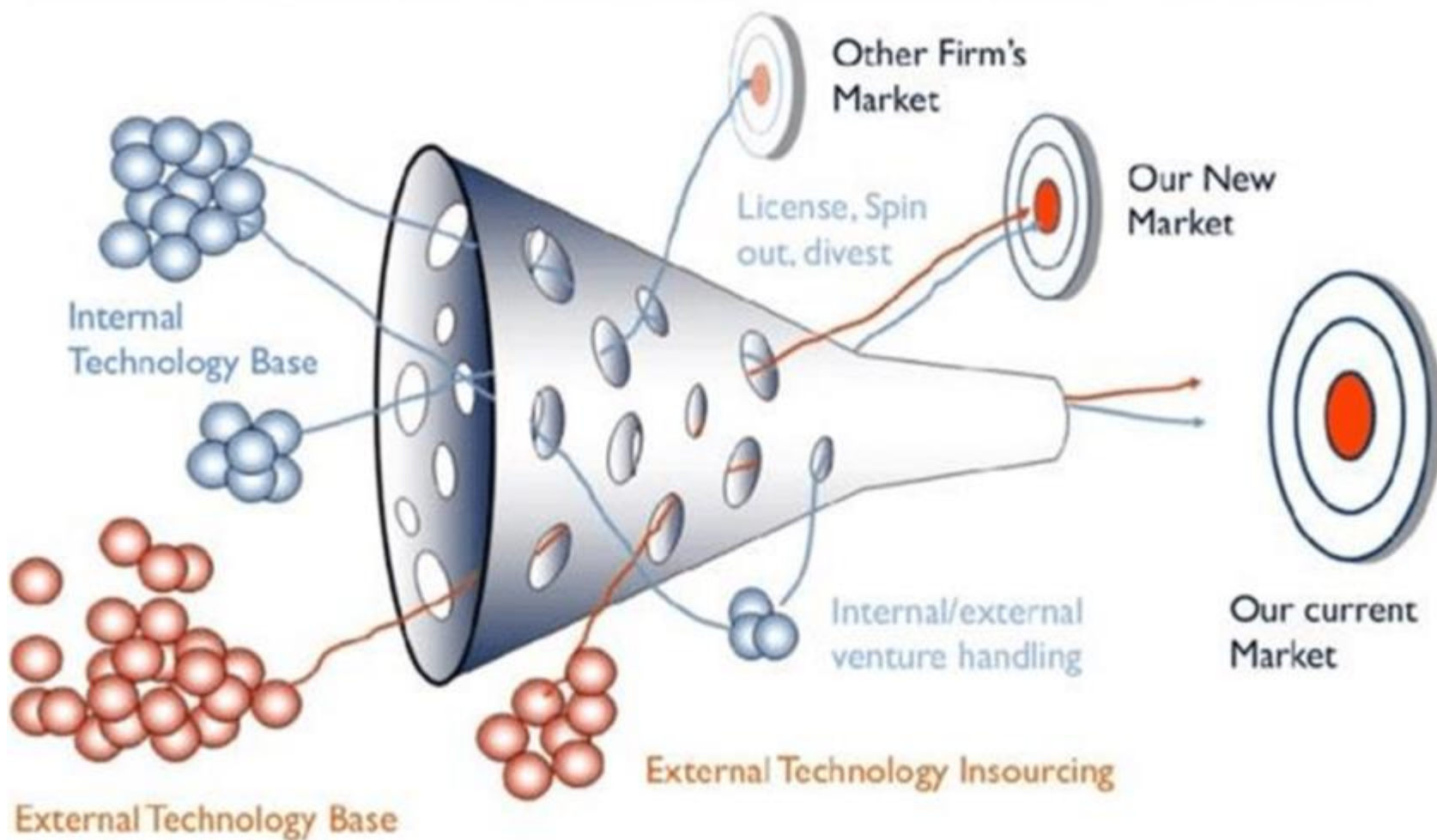


Missoni, Prada, Versace



... and the food industry







- InnoCentive was an open innovation company that crowdsources commission research and development problems.
- It was acquired in 2021 by Wazoku (price 1.3 million £).
- Wazoku is an Enterprise Innovation Platform comprising: Idea Management, Open Innovation Marketplace, Innovation Consulting.



- A world of Unicorns?
- Changing the world... for good!
- No *playground* for old men?

Successful entrepreneurship isn't driven solely by old and experienced companies or young innovators with fresh ideas—it's built by everyone. Entrepreneurship is universal.

- **Where do new firms come from?**

Individual level

- Novice entrepreneur
- Habitual entrepreneur

Organizational level

- **The Entrepreneurial University**
 - Student Entrepreneurship
 - Academic Entrepreneurship
- **Corporate Entrepreneurship (established firm, e.g., family firms)**
 - Corporate spin-off (Employee Entrepreneurship)
 - Intrapreneurship

Social and Public Organizations

- Social entrepreneurship
- Hybrid organizations
- NGO-based ventures

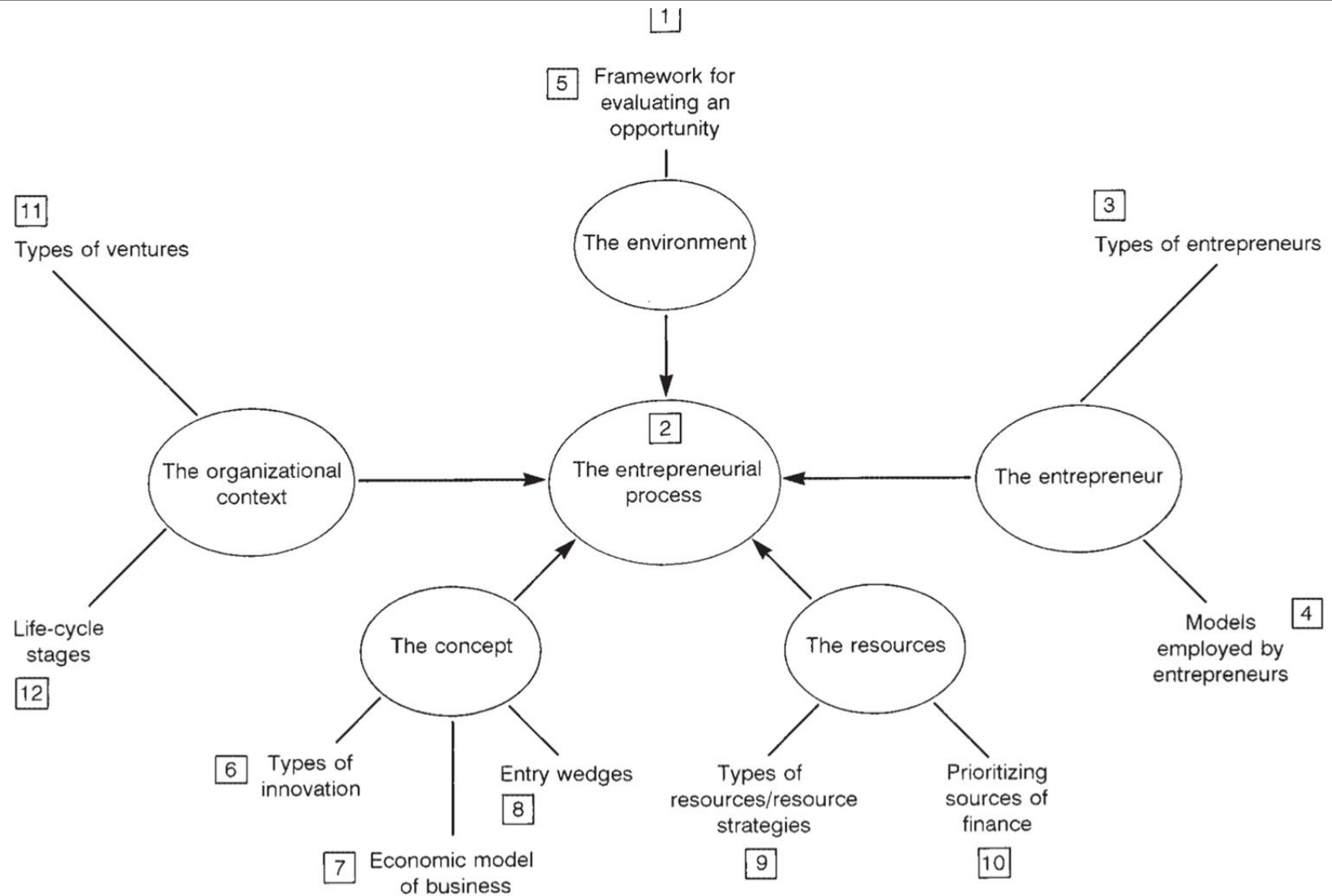
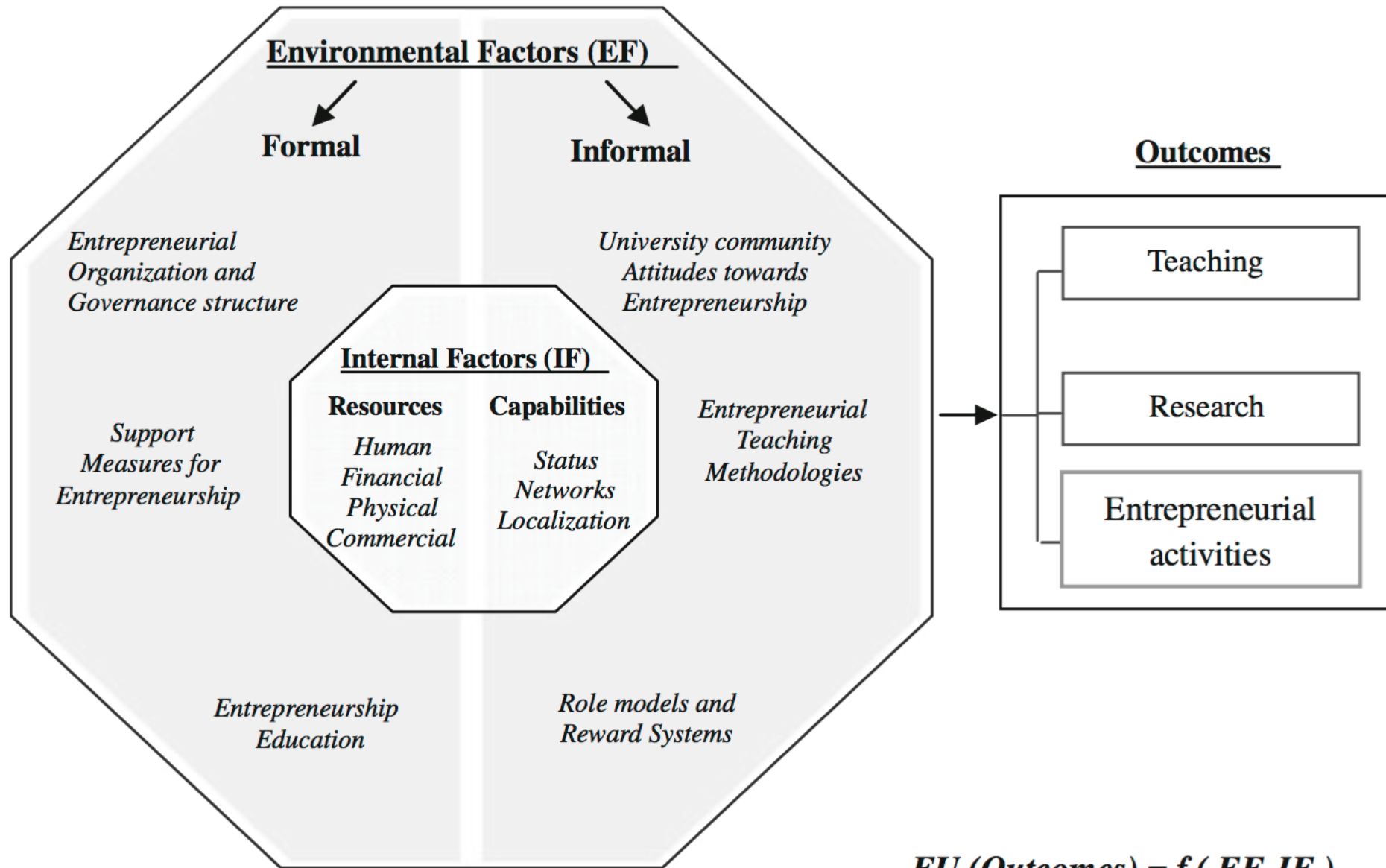


Figure 1. An integrative framework.

- The Knowledge Spillover Theory of Entrepreneurship (KSTE) elucidates how innovation and entrepreneurship are interconnected within a dynamic ecosystem.
 - KSTE posits that knowledge generated within one domain or industry can spill over into adjacent or unrelated domains, sparking entrepreneurial activity.
 - It highlights the role of knowledge diffusion in fostering entrepreneurship and driving economic growth.
- Key Components:
 - Knowledge Creation: Innovations and advancements in technology, science, and business practices contribute to the creation of knowledge.
 - Spillover Mechanisms: Knowledge spillovers occur through various channels, such as collaboration, networking, and mobility of individuals and ideas.
 - Entrepreneurial Response: Entrepreneurs identify opportunities arising from knowledge spillovers and leverage them to create new ventures or innovate existing ones.
- KSTE provides insights into how entrepreneurs can capitalize on knowledge flows and contribute to innovation and growth.

- Guerrero and Urbano (2014) define the entrepreneurial university as an institution that actively promotes a culture of innovation, creativity, and entrepreneurship within its academic community.
- Unlike traditional universities, which primarily focus on teaching and research, entrepreneurial universities actively engage in the commercialization of knowledge and the development of entrepreneurial skills among students and faculty.
- Key dimensions:
 1. **Proactive engagement with industry** and the broader community to identify opportunities for collaboration and knowledge exchange.
 2. **Support for entrepreneurial initiatives and startups**, including the provision of resources, mentorship, and networking opportunities.
 3. **Integration of entrepreneurship education into the curriculum**, equipping students with the skills and mindset necessary to identify and pursue entrepreneurial opportunities.
 4. **A culture of risk-taking and experimentation**, where failure is embraced as a learning opportunity and success is celebrated.
 5. **A commitment to social and economic impact**, with a focus on translating research outcomes into tangible benefits for society.

Conceptual model of entrepreneurial universities (Guerrero and Urbano 2012)



$$EU (Outcomes) = f (EF, IF)$$

World Scientific Series on Public Policy and Technological Innovation – Vol. 1

Student Start-Ups

The New Landscape of Academic Entrepreneurship

Mike Wright • Philippe Mustar • Donald Siegel



 World Scientific



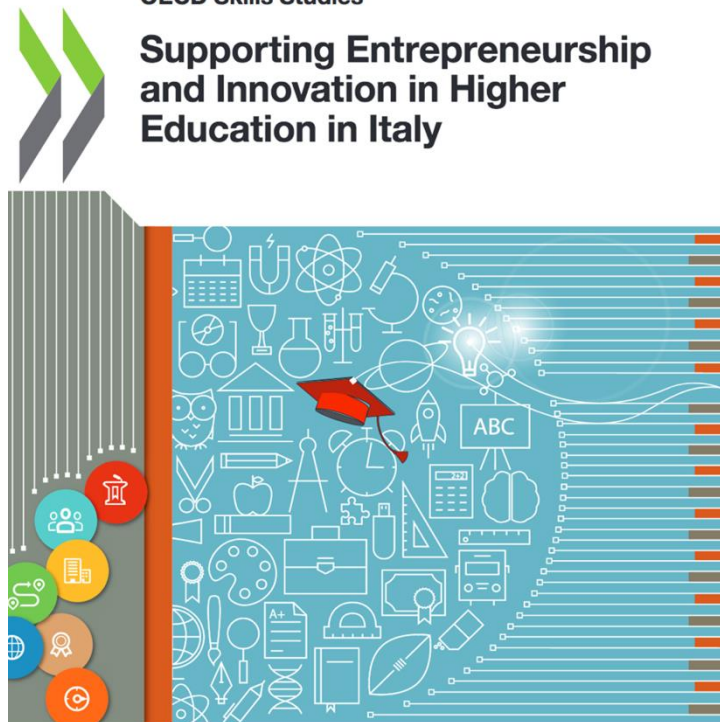
Global University Entrepreneurial Spirit Students' Survey

DEGLI STUDI
DI BERGAMO



OECD Skills Studies

Supporting Entrepreneurship and Innovation in Higher Education in Italy



italian
CLAB NETWORK

Imprenditorialità, Innovazione e Marketing
T. Minola - University of Bergamo

University Entrepreneurship (1) – Student Entrepreneurship

COMPANIES FOUNDED BY GRADUATES

236,362
(2.2%)

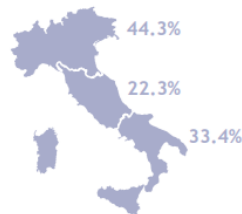
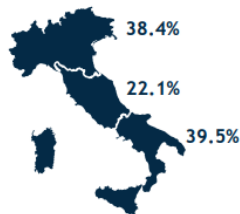
COMPANIES FOUNDED IN ITALY

10,783,746

(Unioncamere Database)

GEOGRAPHICAL AREA

In the ten-year period from 2009 to 2018, 38.4% of companies founded by graduates were headquartered in Northern Italy (compared to 44.3% of all companies founded in Italy), 22.1% in Central Italy (compared to 22.3% of all companies), 39.5% were headquartered in Southern Italy (compared to 33.4% of all companies).



FOUNDERS

7.1%
(205,137)

A founder is a person who owns an equity share and holds a position as administrator or owner in a company when it is founded.



JOINERS

2.3%
(66,098)

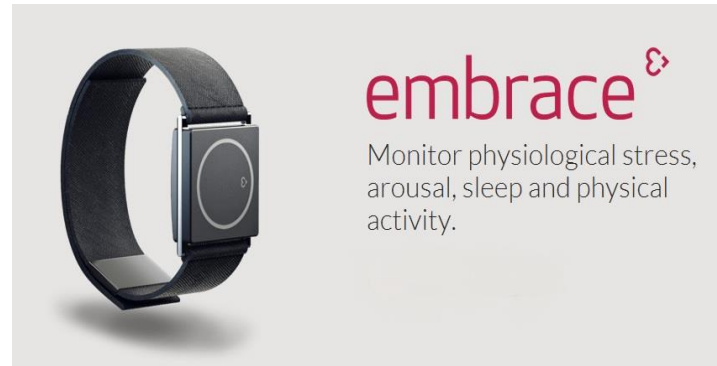
A joiner is a graduate who acquires an equity share in a company equal to at least 10%, after that company was already founded.



POPULATION

2,891,980

Graduates who obtained a degree from an Italian university from 2004 to 2018 (Almalaurea Database)

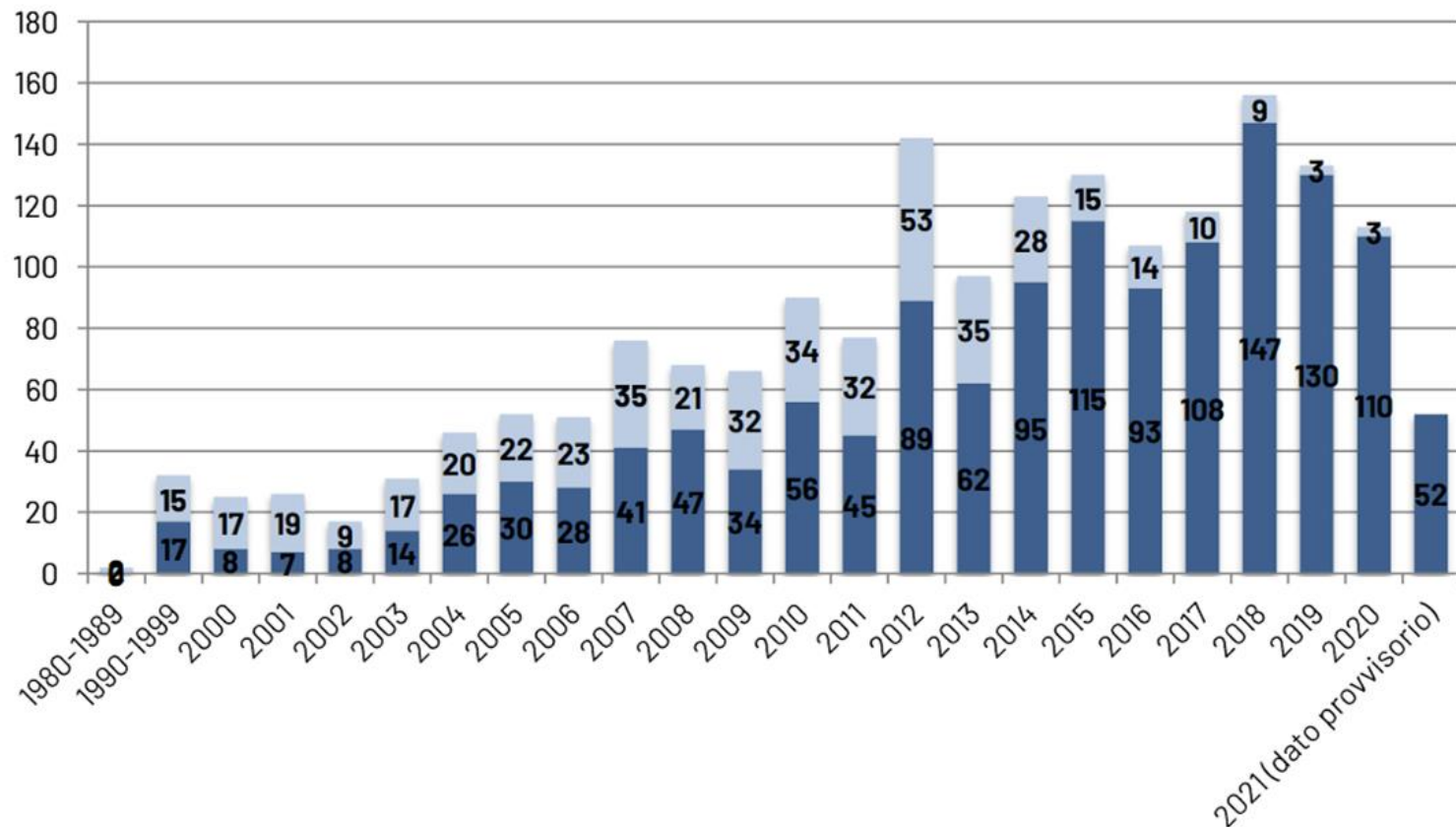


- University spin-offs represent entrepreneurial ventures that emerge from academic institutions with the primary objective of commercializing intellectual property or scientific discoveries originating from university research activities.
- Key Characteristics:
 1. Originating from Academic Institutions: University spin-offs are born out of academic environments, leveraging the knowledge, expertise, and resources available within universities to develop innovative products or services.
 2. Commercialization of Intellectual Property: they focus on translating academic research or intellectual property into commercial applications. This often involves the development of new technologies, products, or services that address market needs or opportunities.
 3. Entrepreneurial Venture: University spin-offs characterized by the pursuit of commercial success and the assumption of risk. They typically operate as independent entities separate from the parent university.

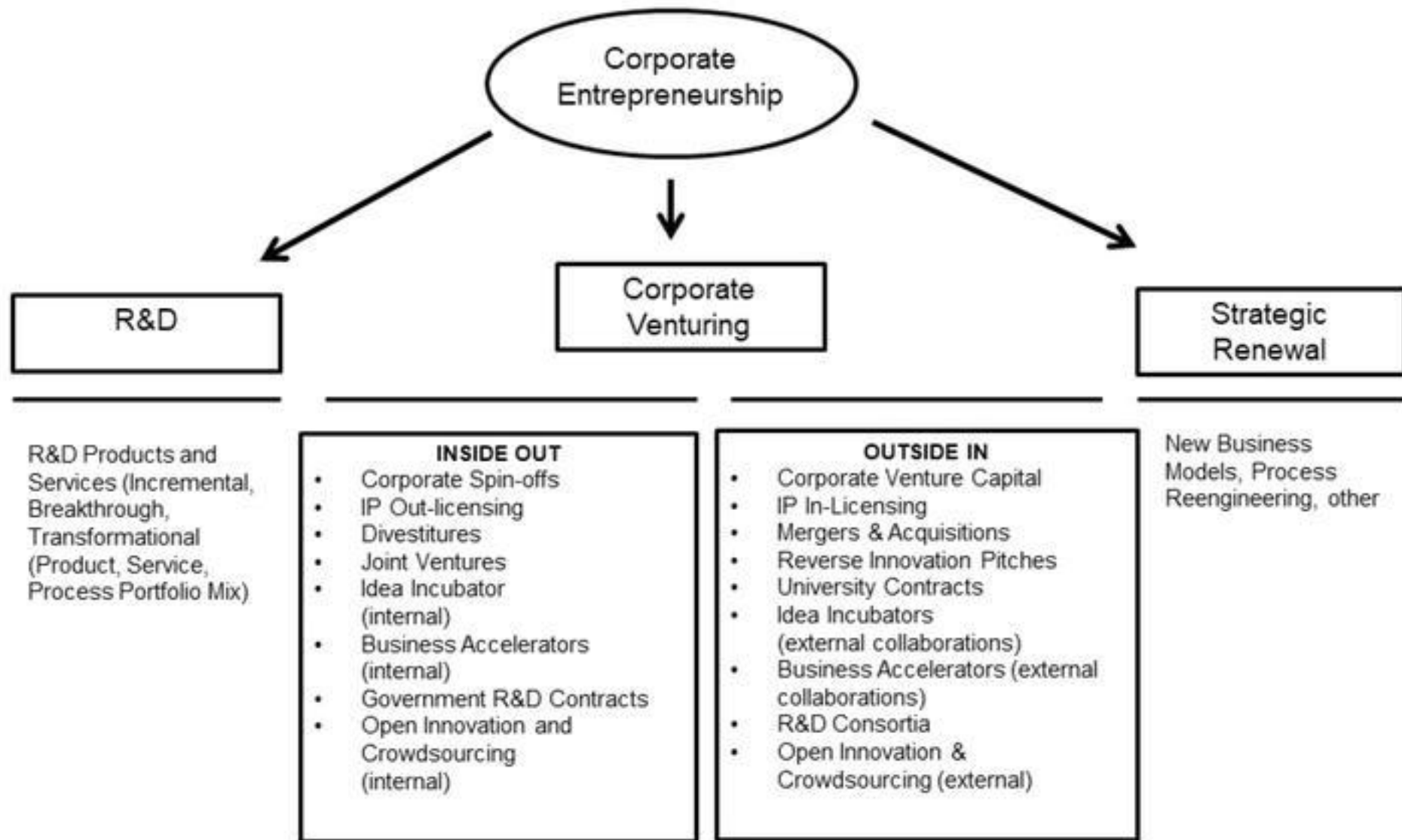


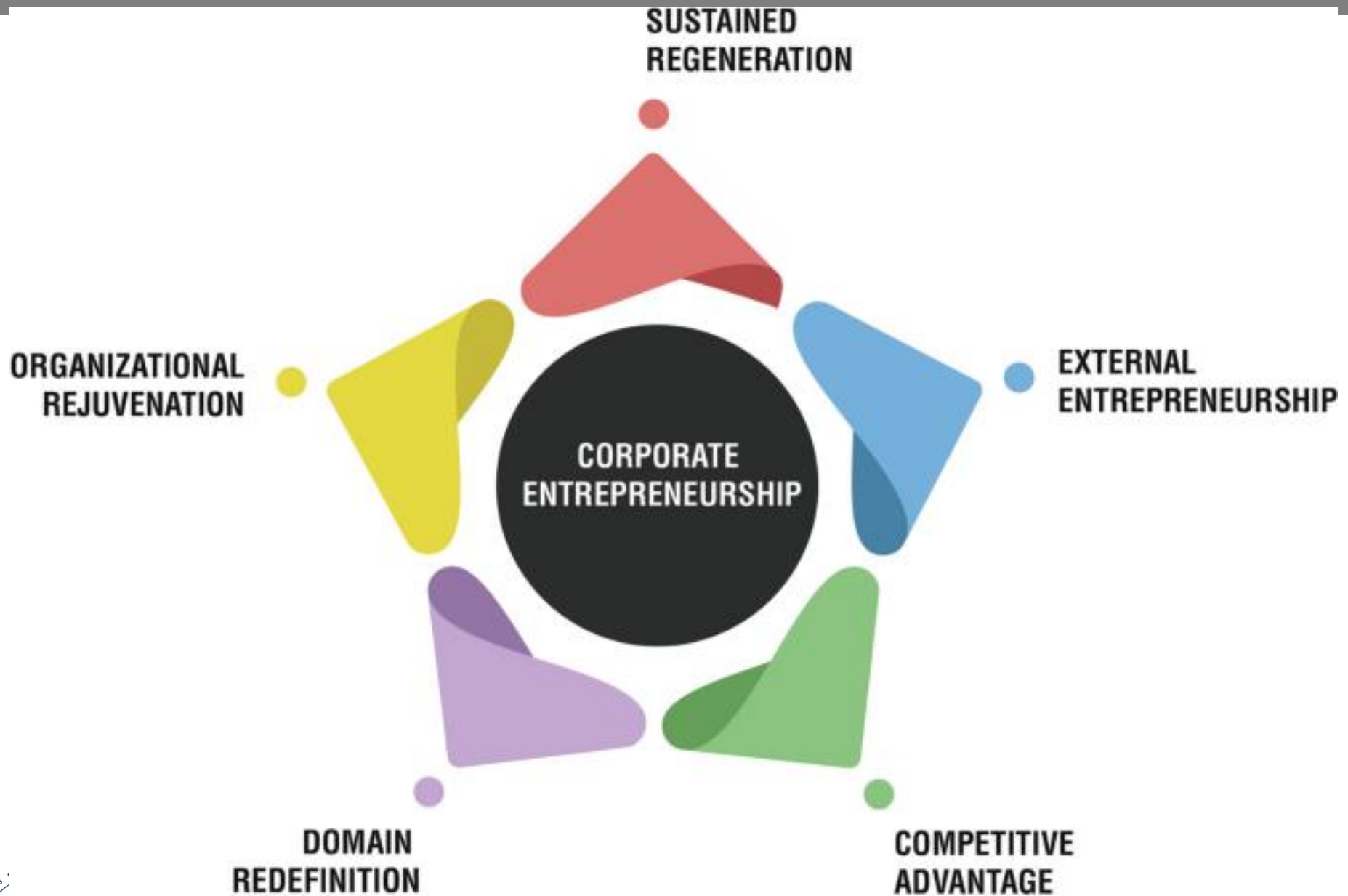


<https://www.kth.se/en/aktuellt/nyheter/cellulose-technology-enables-swift-scaling-up-of-cotton-recycling-enterprise-1.957935>



Academic spin-off in Italy (dark blue = formally affiliated by universities)





- Corporate entrepreneurship refers to the pursuit of entrepreneurial activities within established organizations. It encompasses initiatives aimed at fostering innovation, creating new ventures, and driving organizational renewal and growth from within the corporate context.
- Benefits of Corporate Entrepreneurship:
 - 1. Enhanced Competitiveness: Corporate entrepreneurship enables organizations to stay ahead of the competition by fostering a culture of innovation and driving continuous improvement and renewal.
 - 2. Diversification and Expansion: It allows organizations to diversify their product/service offerings, explore new markets, and expand their business footprint by leveraging entrepreneurial opportunities.
 - 3. Employee Engagement and Retention: Encouraging intrapreneurship fosters employee engagement, creativity, and satisfaction, leading to higher retention rates and a more motivated workforce.
 - 4. Future-proofing: By embracing corporate entrepreneurship, organizations can adapt to disruptive changes in the business environment, anticipate future trends, and position themselves for long-term success.



UPDATED

Version 6.8

EasyWOOD

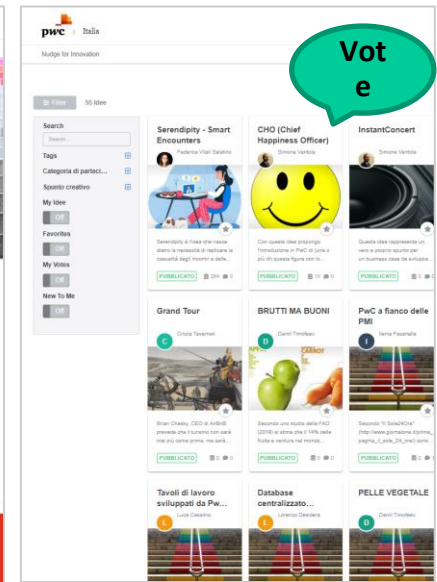
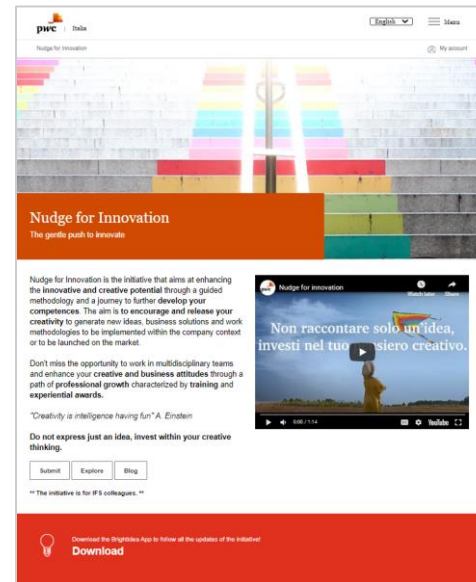
DDX Software Solutions | info@ddxgroup.com | www.ddxgroup.com



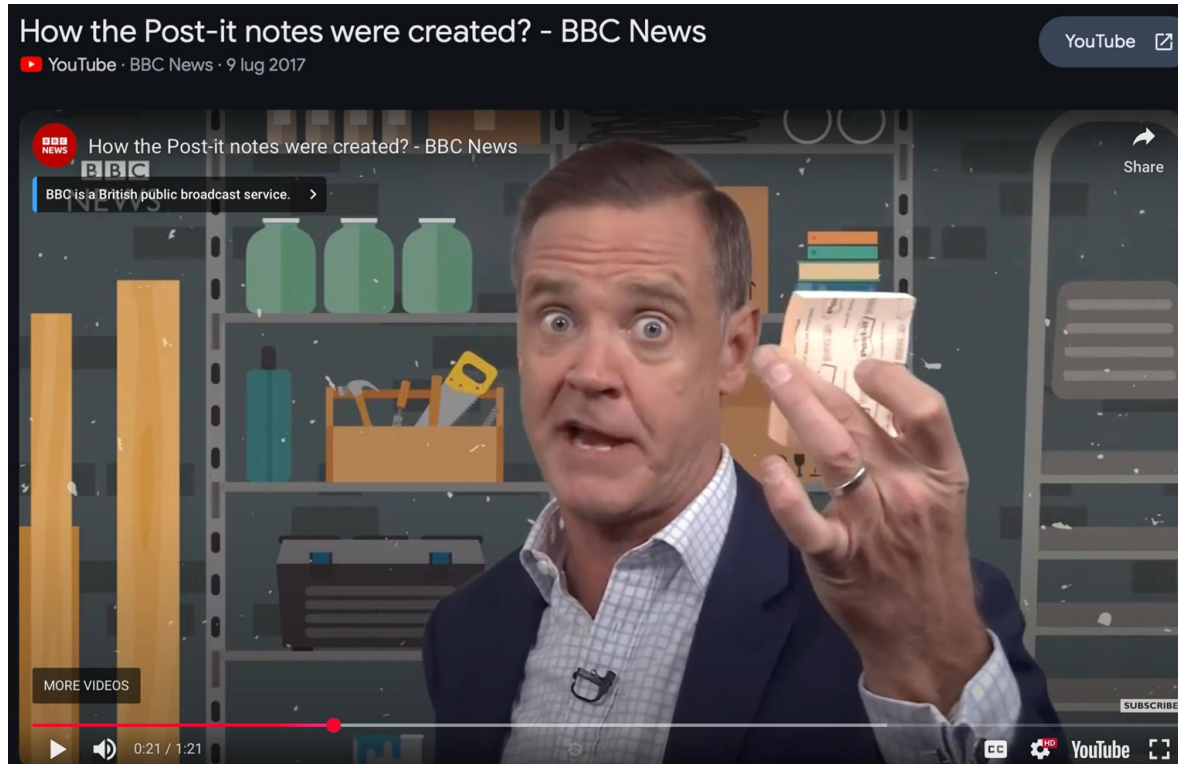
- Intrapreneurship refers to the practice of fostering entrepreneurial behavior and innovation within an existing organization, typically by empowering employees to explore new ideas, take risks, and pursue opportunities for growth and improvement.
- Key Elements of Intrapreneurship:
 - 1. Entrepreneurial Mindset
 - 2. Risk-Taking
 - 3. Resource Allocation
 - 4. Organizational Support
- Benefits of Intrapreneurship:
 - 1. Innovation
 - 2. Employee Engagement
 - 3. Organizational Growth
 - 4. Competitive Advantage

Why is entrepreneurship important for employees and employers?

- **Nudge for Innovation: a gentle push towards innovation**



Brightidea for Idea Generation & Innovation Management Workflow



https://youtu.be/U2Lww_W-9qc?feature=shared

3M Post-it Notes

- Intrapreneurship Story: Spencer Silver developed a weak adhesive at 3M, which initially seemed useless. However, Art Fry, another employee, recognized its potential for reusable sticky notes. Despite initial skepticism, Post-it Notes became a billion-dollar product, generating over \$1 billion annually.
- Impact: This accidental innovation turned into a ubiquitous office supply item, showcasing how intrapreneurship can transform seemingly failed projects into successes.



- Intrapreneurship Story: Amazon Prime was not born from a corporate strategy meeting but from an intrapreneurial idea by Charlie Ward, an Amazon engineer. He recognized that customers wanted faster delivery without additional shipping costs. Despite initial skepticism from Jeff Bezos, Ward's idea was given the green light, and Amazon Prime launched in 2005. Today, it generates over \$19 billion annually with more than 200 million subscribers worldwide.
- Impact: This initiative transformed customer expectations for fast shipping and became a cornerstone of Amazon's success.

- A world of Unicorns?
- Changing the world... for good!
- No *playground* for old men?
- Where do new firms come from?

Entrepreneurship can arise from university students (Student Entrepreneurship), faculty (Academic Entrepreneurship), or within established companies (Corporate Entrepreneurship). Being an entrepreneur doesn't always mean starting your own business—you can also innovate from within an organization as an intrapreneur to create real impact.

- **A new era...**

- Approximately 95% of all businesses are small
- 70% of the labor force is employed by small businesses
- More and more business and managerial engineering graduates are attracted to the demands of small business and the potential payoffs
- We have entered the **Entrepreneurial Society Era**

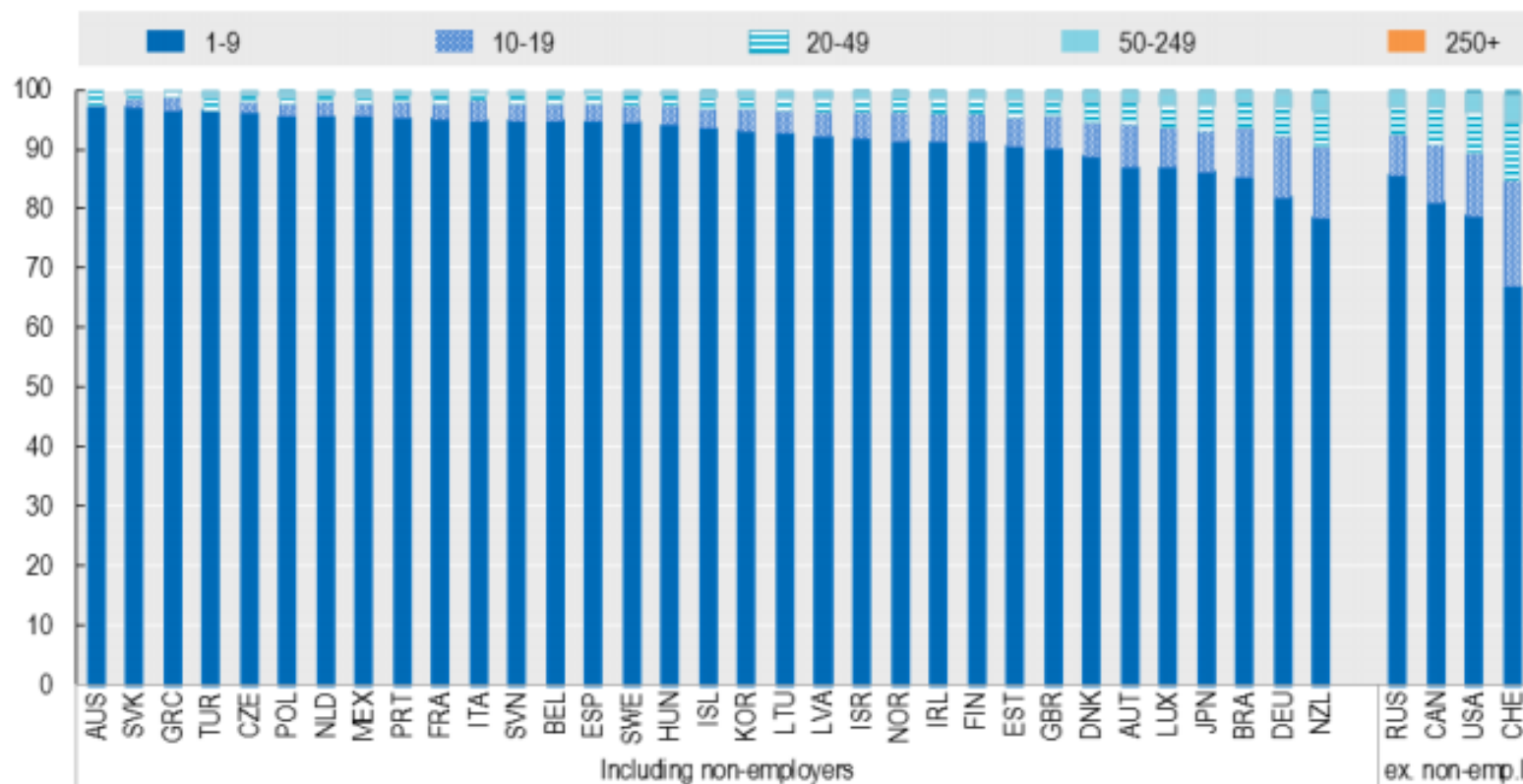
- Innovation comes from the ability to be **flexible** and **adaptable** to technological changes in the environment (Apple, Kodak)
- Economies of scale can be obtained **even by smaller businesses**: flexible manufacturing, outsourcing, strategic alliances (e.g. 3D printing)
- Small business can attract **innovators**

- The business is managed by the owner or owners of the firm
- Decision making lies mostly with one executive
- Annual sales are less than € 10 million
- The business employs less than 50 people
- There is a close management-employee relationship
- Normally the company stock is not listed with a stock exchange
- The firm is not dominant in the industry (size of business, market share)

Enterprise category	Headcount	Turnover or	Balance sheet total
medium-sized	< 250	< € 50 million	< € 43 million
small	< 50	< € 10 million	< € 10 million
micro	< 10	< € 2 million	< € 2 million

6. Number of enterprises by size, business economy

Percentage of all enterprises, 2016, or latest available year



- Radical and rapid changes in new technologies, such as ICT, nanotechnology and renewable energies, and new opportunities coming from the globalized economy have created market chances that are more and more developed by new firms or small firms than by established companies.
- So, while in the past economic studies pointed to the importance of large firms, now a key shift has occurred to the **small and medium enterprises** (SMEs) and the relevant **role of entrepreneurship and entrepreneurs**.
- The course aims at analyzing the entrepreneurial phenomena from different perspectives, both individual and corporate level, at their early stage (start-up) but also in the case of established firms, such as family businesses. A special focus is more and more given to the role of **young entrepreneurs**.

Entrepreneurs Will Create the Future [Entire Talk]

Reid Hoffman, LinkedIn

Entrepreneurial Thought Leaders / June 29, 2011

43 MINUTES

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In this keynote address from the Endeavor Entrepreneur Summit, serial entrepreneur and LinkedIn Co-Founder Reid Hoffman shares his views on the role of entrepreneurs in society and the impact they will play in the creating the future. Hoffman offers valuable rules of thumb for navigating the startup and entrepreneurial process, as well

<https://ecorner.stanford.edu/videos/entrepreneurs-will-create-the-future-entire-talk/>

A change in competitive pattern

- Used to be: Large vs. Small
- Has turned to be: Fast vs. Slow



- Previous generations enjoyed the security of lifelong employment with a sole employer. Public policy and social institutions reinforced that security by producing a labor force content with mechanized repetition in manufacturing plants, and creating loyalty to one employer for life.
- This is no longer the case. Globalization and new technologies have triggered a shift away from capital and towards knowledge. In today's global economy, where jobs and factories can be moved quickly to low-cost locations, **the competitive advantage has shifted to ideas, insights, and innovation.**
- But it is not enough just to have new ideas. It takes entrepreneurs to **actualize them** by championing them to society.
Entrepreneurship has emerged as the proactive response to globalization.

David B. Audretsch, The Entrepreneurial Society



	Resource Era	Industrial Era	Knowledge Era	Entrepreneurial Era
Main industries	Mining, steel and ship building	Automobile and oil	Computer, Telecommunications and Service	Emerging Technologies and Niches
Key assets	Mills and shipyards	Factories and Equipment	Knowledge and knowledge workers	Entrepreneurial approach
Speed of Innovation	Low	Medium	High	Very high continuous
Size of Operation	Large	Large	Small independent business units	Small
Business Cycle	Long	Long to Medium	Medium to Short	Short

- IT technologies for adding value to information
- Synthetic biology and metabolic engineering
- Green Revolution 2.0 – technologies for increased food and biomass
- Nanoscale design of materials
- High energy density power systems
- Personalized medicine, nutrition and disease prevention
- Distributed Storage
- Wireless power
- Power Grid Control
- (...)

- Under the managed economy of the cold war era, government policies around the world supported big business, while small business was deemed irrelevant and largely ignored. Now a fundamental policy revolution is underway, shifting the focus to technology and knowledge-based entrepreneurship, where **start-ups and small business have emerged as the driving force of innovation, jobs creation, competitiveness and growth.**
- The role of the university has accordingly shifted from tangential to a highly valued seedbed for coveted new ideas with the potential to create not just breathtaking new ventures, but also entire new industries.
- By understanding the **shift from the managed economy and the emergence of the entrepreneurial society**, individuals, businesses, and communities can learn how to proactively harness the opportunities afforded by globalization in this new entrepreneurial society.

David B. Audretsch, The Entrepreneurial Society

- A world of Unicorns?
- Changing the world... for good!
- No *playground* for old men?
- Where do new firms come from?
- A new era...

The economy has evolved from being resource-based to knowledge-driven. This entrepreneurial era is demanding entrepreneurs to tackle complex challenges and drive substantial socioeconomic growth through the development and use of **new technologies**.

- **Digitization vs. Digitalization vs. Digital transformation**
- It's all about human being (and being human)...

Digitization is the straightforward process of converting analog information to digital—turning pages into bytes, for instance, by scanning a document or uploading a sound recording. It often also captures the process of moving a process from manual to digital—replacing hand-filled forms with online versions that go directly into a database, for instance. The much talked-about, and ever elusive, “paperless office” is the pinnacle of digitization.

Digitalization refers to the use of digital technology, and probably digitized information, to create and harvest value in new ways.

What digitization doesn’t do is change the business itself; it doesn’t drive new business models or disrupt fundamental business strategies. That is the domain of digitalization.

Digital transformation is the use of new digital technologies such as social media, mobile technology, analytics, or embedded devices to enable major business improvements including enhanced customer experiences, streamlined operations, or new business models.

AI vs. Gen AI vs. Machine Learning vs. LLMs

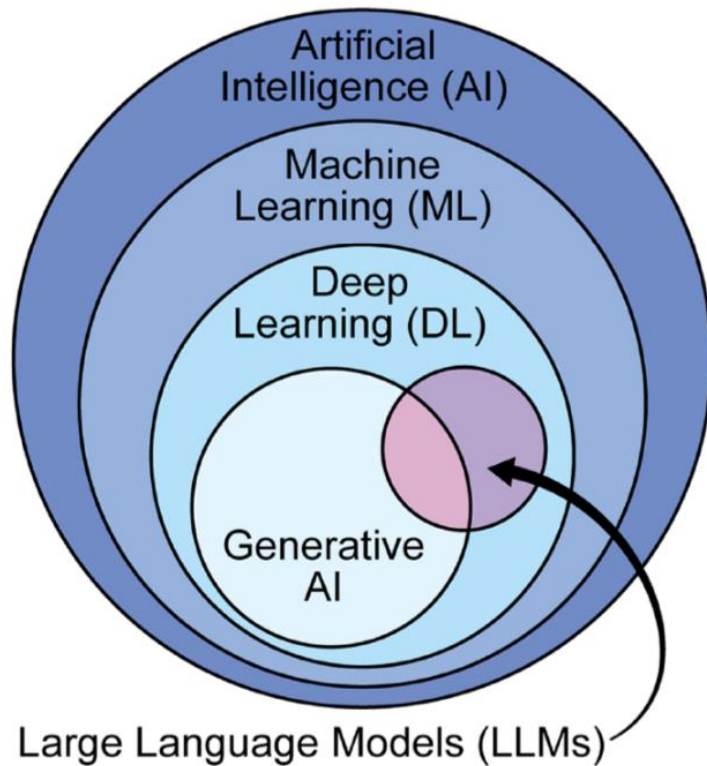
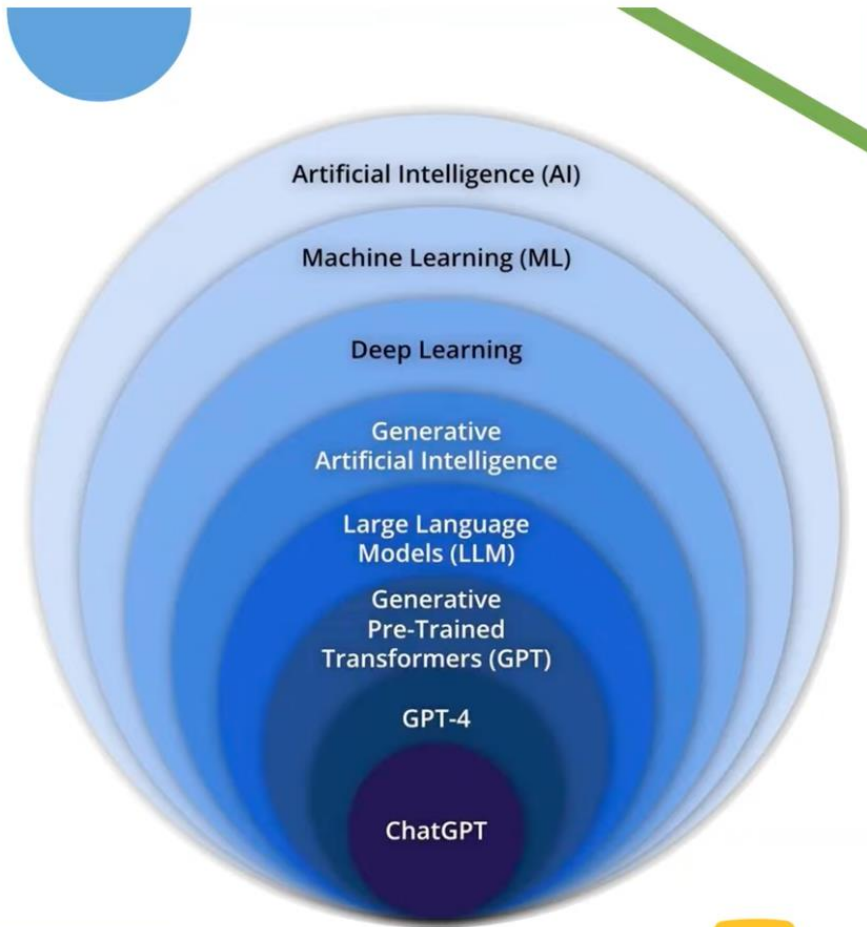


Figure 1. LLMs within the AI taxonomy. LLMs exist as a subset of deep learning models, which are a subset of machine learning models. Machine learning models are a subset of AI. Depending on their size, LLMs can also be considered generative AI models. AI, artificial intelligence; LLM, large language model.



AI: a technology that enables machines to imitate various complex human skills.

- **Machine Learning**

- Supervised Learning (Classification, Regression)
- Unsupervised Learning (K-means)
- Semi-supervised Learning
- Reinforcement Learning
- **Deep Learning (large dataset, dominant)**
 - **Generative AI** : generative modeling that is instantiated with a machine learning architecture (e.g., a deep neural network) and, therefore, can create new data samples based on learned patterns.

- **LLM** (for Natural Language Processing (NLP) tasks)

=> Transformer language models that contain hundreds of billions (or more) of parameters , which are trained on massive text data.

=> LLMs exhibit strong capacities to understand natural language and solve complex tasks (via text generation).

Sheikh, H., Prins, C., & Schrijvers, E. (2023). Artificial intelligence: definition and background. In *Mission AI: The new system technology* (pp. 15-41). Cham: Springer International Publishing.

Feuerriegel, S., Hartmann, J., Janiesch, C. et al. Generative AI. *Bus Inf Syst Eng* 66, 111–126 (2024). <https://doi.org/10.1007/s12599-023-00834-7>

Zhao, Wayne Xin, Kun Zhou, Junyi Li, Tianyi Tang, Xiaolei Wang, Yupeng Hou, Yingqian Min et al. "A survey of large language models." *arXiv preprint arXiv:2303.18223* 1, no. 2 (2023).

- A world of Unicorns?
- Changing the world... for good!
- No *playground* for old men?
- Where do new firms come from?
- A new era...
- Digitization vs. Digitalization vs. Digital transformation
Digitization converts analog information into digital form; digitalization uses digital technologies to create new value; and digital transformation leverages advanced digital technologies to fundamentally change business models and organizational processes.
- **It's all about human being (and being human)...**

"The three things entrepreneurs do for our economy"

Kauffman Sketchbook

America and the World
will as long as human beings
walk this planet
need:



<https://youtu.be/M7VZIbeUrSU>

When talents meet opportunity:

A successful startup

January, 2009:

Jan Koum bought an iPhone

He realized that Apple's 7-month-old App Store could be an opportunity

He started developing an app with **Brian Acton**
They decided to open their first startup: **Whatsapp inc.**



Their app kept crashing or getting stuck
Jan told Brian to search for a new job



January, 2009

2009 May, 23rd:
Brian was turned down by Twitter



Second fail

January, 2009

2009 May, 23rd

2009 Aug, 3rd:
Brian was turned down by Facebook



Third fail

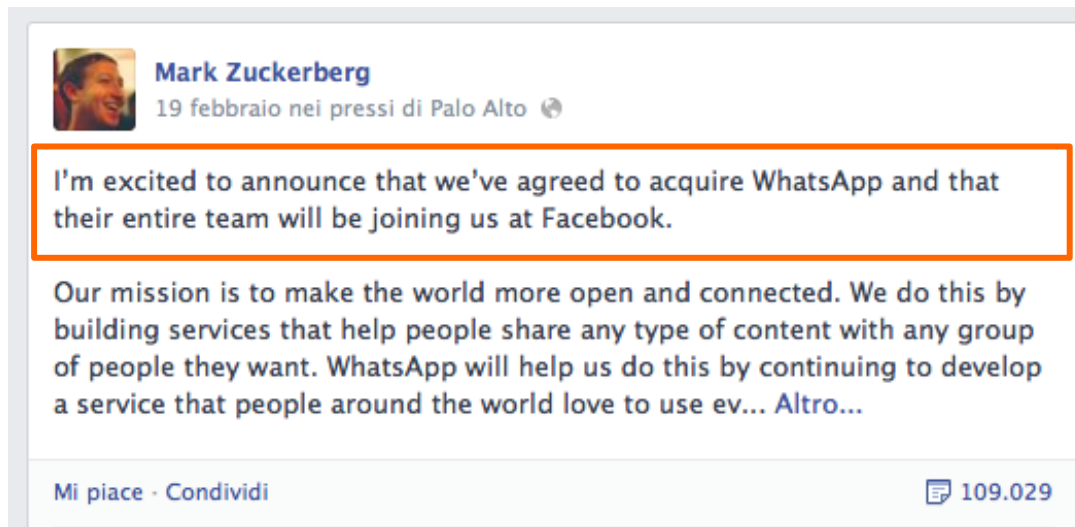
January, 2009

2009 May, 23rd

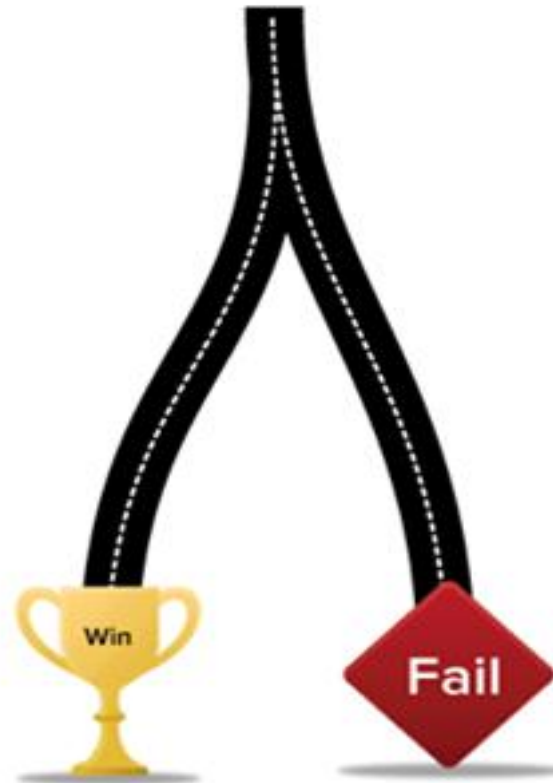
2009 Aug, 3rd



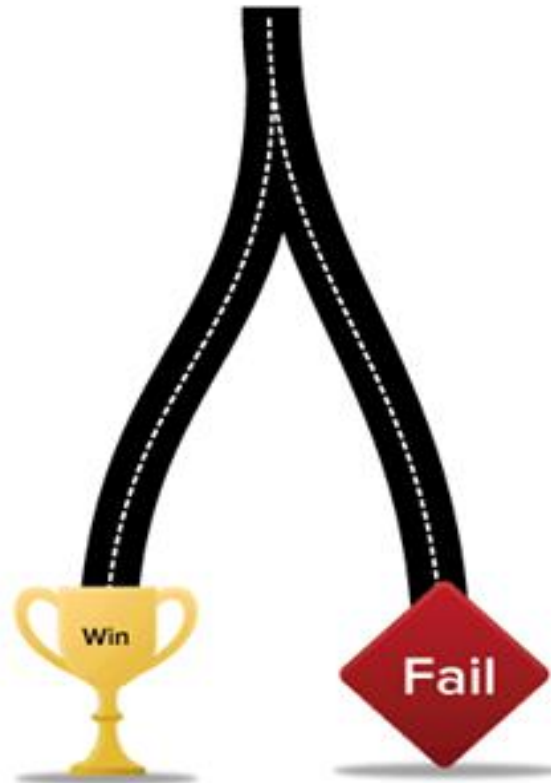
2014 Feb, 19th:
Facebook buys Whatsapp for **\$19 billions**



What Most People Think



What Most People Think



What Successful People Know



Why we study **Entrepreneurship** *(as Engineers)*

Why we study Entrepreneurship



<https://www.youtube.com/watch?v=WZHvRpuemgk>

“Do you want to sell sugared water for the rest of your life? Or do you want to come with me and change the world? “

Steve Jobs to John Sculley (president of PepsiCo), 1983

Why we teach Entrepreneurship

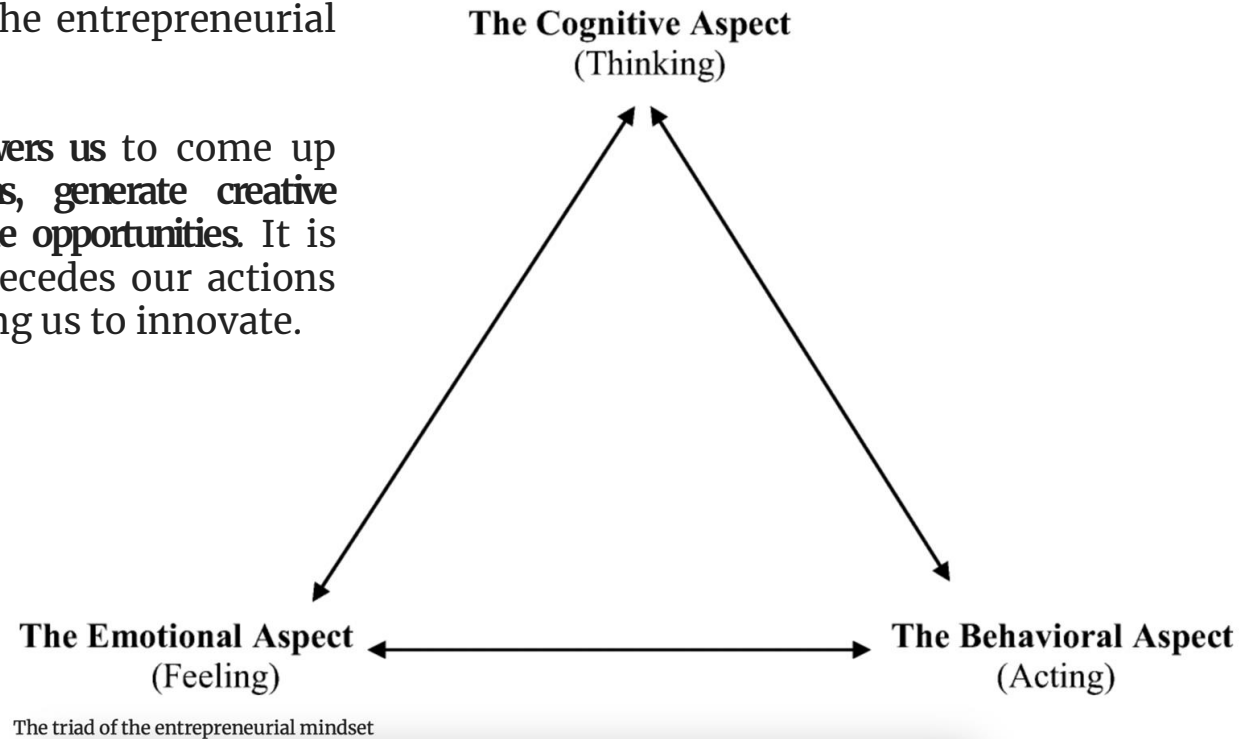
"Entrepreneurial Mindset"

Kauffman Sketchbook



<https://www.youtube.com/watch?v=8NBnoVrLFPU>

- The **true source of innovation and entrepreneurship** is an **ability and perspective** that resides within each one of us, something we refer to as the entrepreneurial mindset.
- This mindset **allows and empowers** us to come up with **new ideas, solve problems, generate creative solutions, and take action to pursue opportunities**. It is the mental perspective that precedes our actions and feeds our emotions, allowing us to innovate.



1. The cognitive aspect—how entrepreneurs use mental models to think.
2. The behavioral aspect—how entrepreneurs engage or act for opportunities.
3. The emotional aspects—what entrepreneurs feel in entrepreneurship.

What is entrepreneurship education?

- **Entrepreneurship** is all about creating value, not only in terms of financial returns, but also about improving social and environmental aspects.
- **Entrepreneurship education** prepares people to be responsible and enterprising individuals. It helps people develop the skills, knowledge, and attitudes necessary to achieve the goals they set out for themselves. Evidence also shows that people with entrepreneurial education are more employable.
- Young people with entrepreneurship education are more likely to set up their own companies. Up to 20% of students who participate in a mini-company program in secondary school will later start their own company. That is up to five times higher than in the general population.
- **Entrepreneurship can be learnt.** You don't have to be born as “entrepreneur” to launch and run a successful business. You can become one by developing an **entrepreneurial mindset and skills.**



Source : European Commission > Entrepreneurship and Small and medium-sized enterprises (SMEs) > Entrepreneurship education

Entrepreneurial Skills Learned

Mark Zuckerberg, Facebook

Entrepreneurial Thought Leaders / October 26, 2005

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0:08 / 2:07

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Zuckerberg talks about the entrepreneurial skills he acquired in college that led him to building Facebook.

<https://ecorner.stanford.edu/videos/entrepreneurial-skills-learned/>

- A world of Unicorns?
- Changing the world... for good!
- No *playground* for old men?
- Where do new firms come from?
- A new era...
- It's all about human being (and being human)...

Entrepreneurship requires people who, with the right training, can develop the skills and competencies to act and become entrepreneurs. However, taking the leap starts with building an entrepreneurial mindset.

- **Entrepreneurship: an art more than a science, or not?**

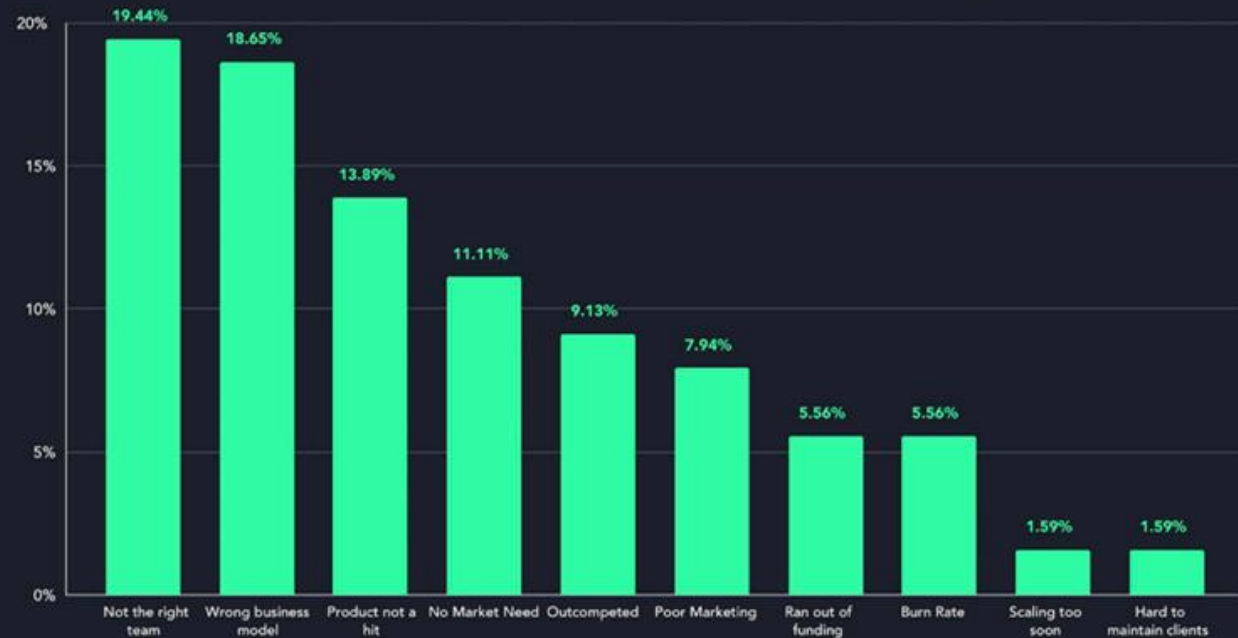
AUTOPSY: TRANSFORMING FAILURE INTO SUCCESS

# Within 3 years 92% of startups fail. Learn from their mistakes.

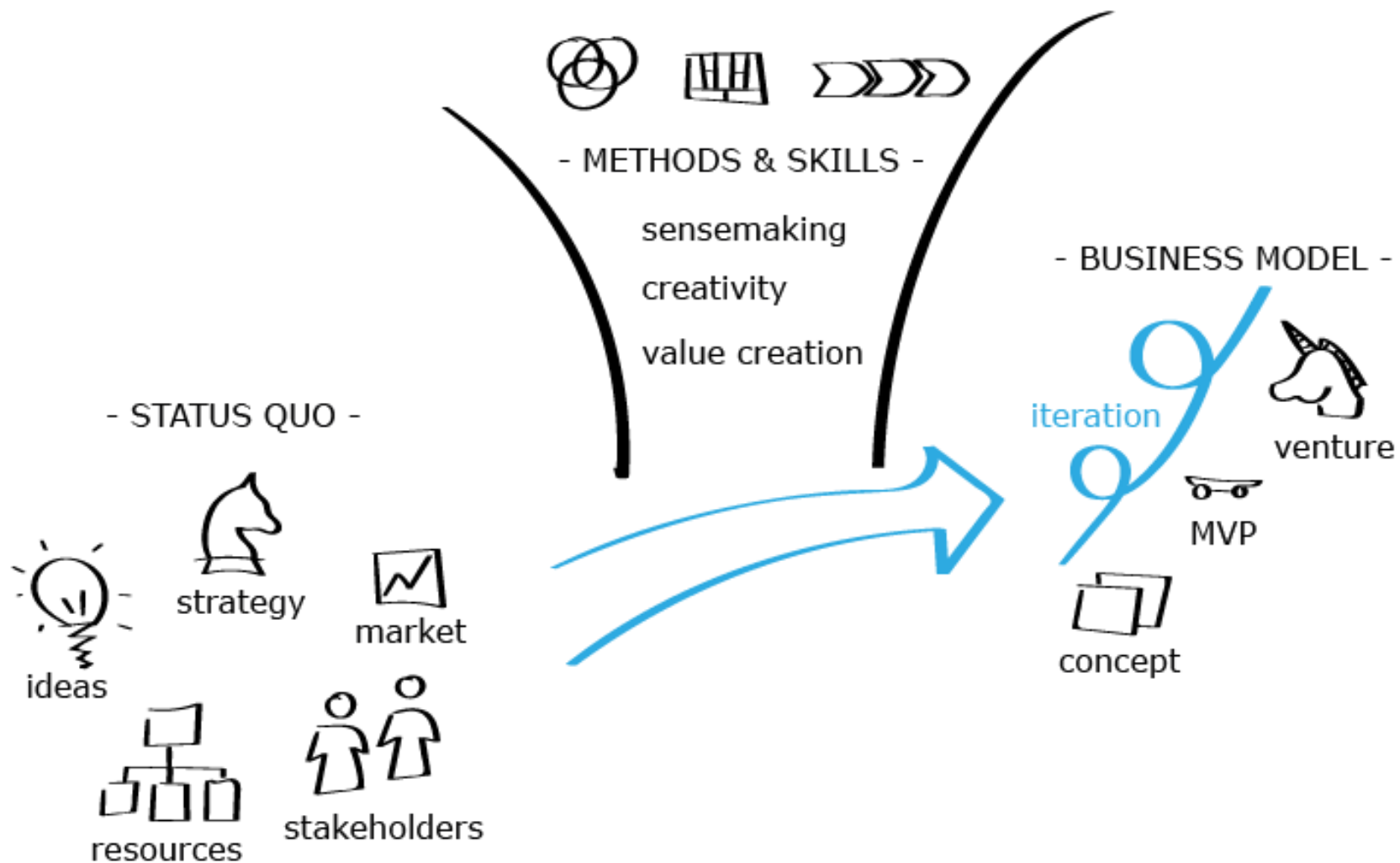
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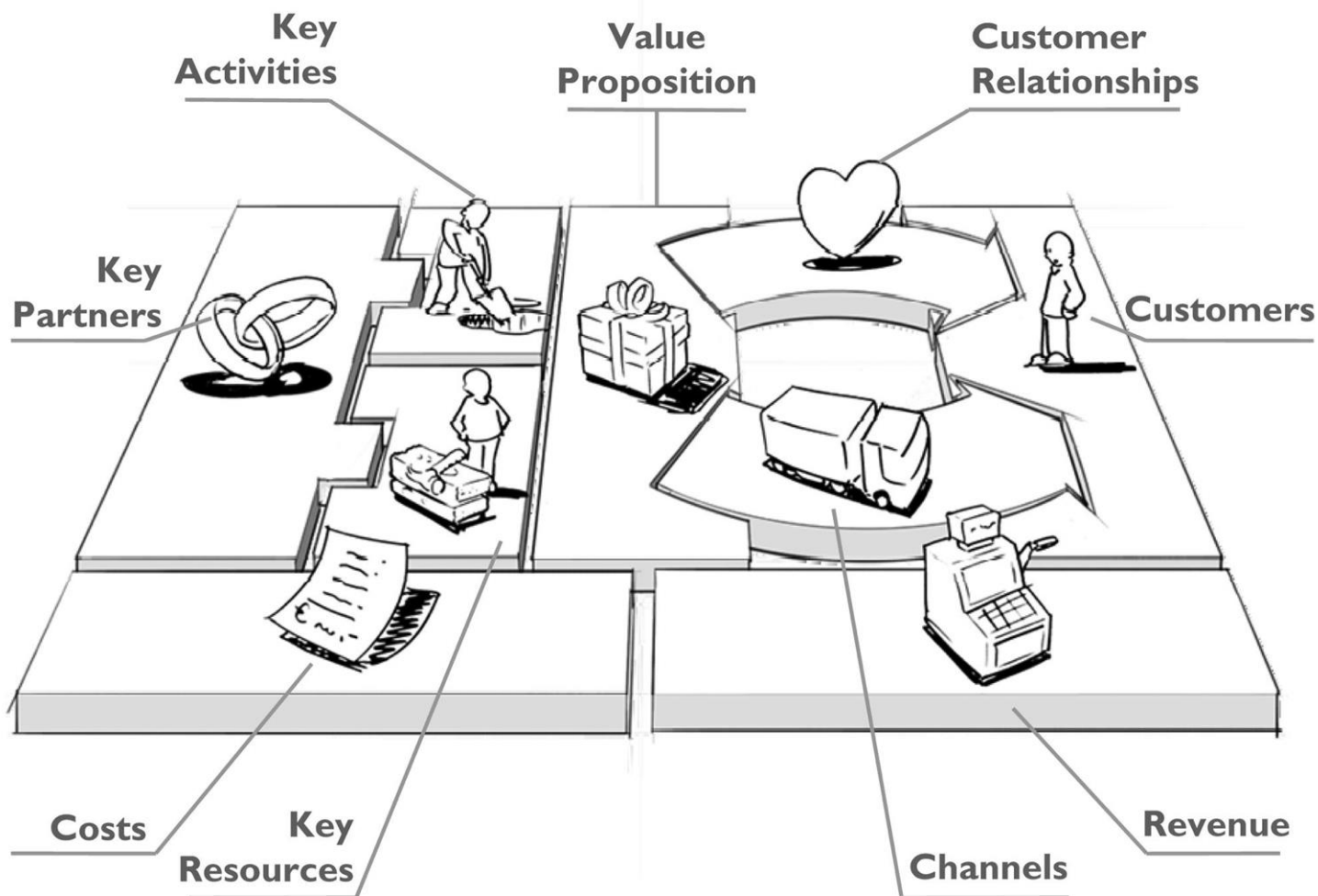
JOIN 10,000+ AUTOPSY

**Autopsy Data Analysis: Top 10 reasons startups fail based on 300 failed startup autopsies**

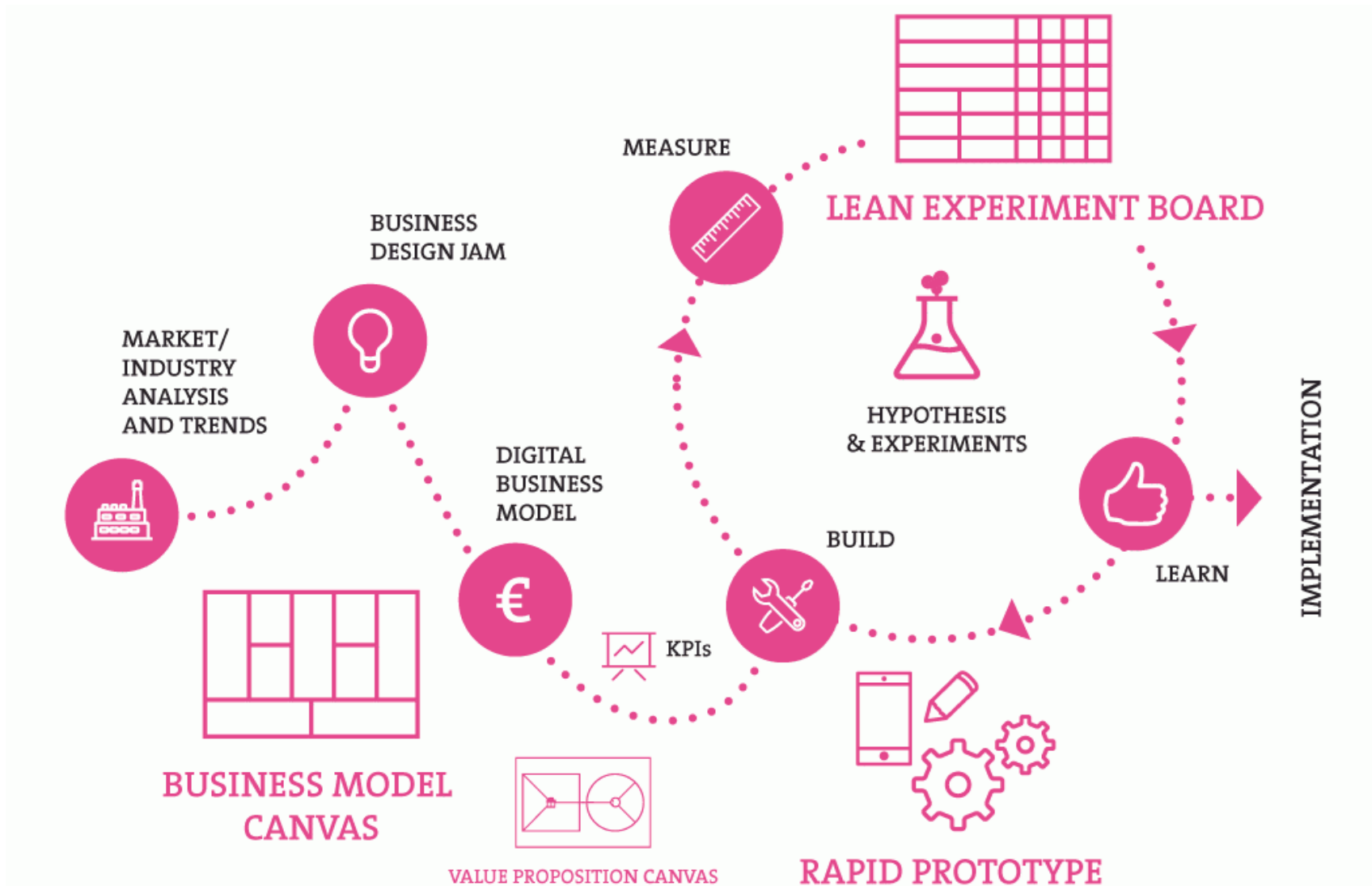


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drawings by JAM





- A world of Unicorns?
- Changing the world... for good!
- No *playground* for old men?
- Digital Entrepreneurship (health, AI, platform economy)
- Where do new firms come from?
- A new era...
- It's all about human being (and being human)...
- Entrepreneurship: an art more than a science, or not?

Entrepreneurship comes with uncertainty, but the right strategies and resources can make the journey smoother. In this course, we'll equip you with essential theories and practical tools to navigate the challenges effectively.

---

**TOMMASO MINOLA**

Department of Management, Information and Production Engineering  
*and* Center for Young and Family Enterprise  
*University of Bergamo*

**MARTA CACCAMO**

Media, Management and Transformation Centre  
*Jönköping International Business School*

IMPRENDITORIALITÀ, INNOVAZIONE E MARKETING

## Course Introduction



**UNIVERSITÀ  
DEGLI STUDI  
DI BERGAMO**